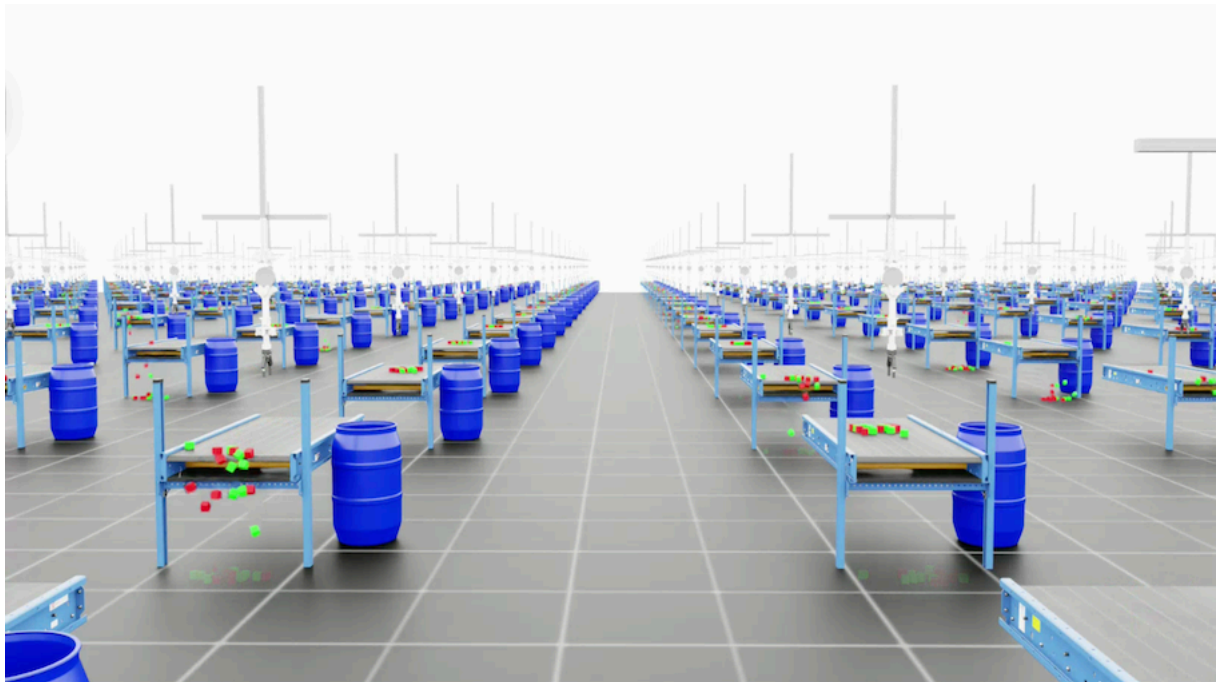


Simulative Construction and Verification of a Tendon-Driven Robot for Conveyor-Based Pick-and-Place Tasks Using Isaac Lab

Project Thesis in the program Computational Engineering M. Sc.

Friedrich-Alexander-Universität Erlangen-Nürnberg
Lehrstuhl für Fertigungsautomatisierung und Produktionssystematik
Prof. Dr.-Ing. J. Franke



Author: Georg Meyer

Student ID: 22791103

Supervisors:

Deadline: 01.07.2026

Duration: x months

Erklärung

Ich versichere, dass ich die vorliegende Arbeit ohne fremde Hilfe und ohne Benutzung anderer als der angegebenen Quellen angefertigt habe und dass die Arbeit in gleicher oder ähnlicher Form noch keiner anderen Prüfungsbehörde vorgelegen hat und von dieser als Teil einer Prüfungsleistung angenommen wurde. Alle Ausführungen, die wörtlich oder sinngemäß übernommen wurden, sind als solche gekennzeichnet.

Erlangen, den 02.05.2026

Georg Meyer

Table of Contents

Erklärung	3
Guideline	1
0.1 Format and Style	1
0.1.1 Page Layout and Typography	1
0.1.2 Page Numbering and Binding	1
0.1.3 Section and Chapter Formatting	1
0.1.4 Paragraphs, Spacing, and Whitespace	1
0.1.5 Figures	2
0.1.6 Tables	2
0.1.7 Diagrams	2
0.1.8 Formulas	3
0.1.9 Appendix	3
0.1.10 Eidesstattliche Erklärung (Statutory Declaration)	3
0.2 Thesis Structure and Content	3
0.2.1 Required Document Order	3
0.2.2 Page Count Guidelines	3
0.2.3 Red Thread (Roter Faden)	4
0.2.4 Chapter: Einleitung und Zielstellung (Introduction)	4
0.2.5 Chapter: Grundlagen und Stand der Technik (Background and State of the Art)	4
0.2.6 Chapter: Handlungsbedarf (Research Gap)	4
0.2.7 Chapter: Methodenteil (Methodology)	5
0.2.8 Chapter: Ergebnisteil (Results)	5
0.2.9 Chapter: Diskussionsteil (Discussion)	5
0.2.10 Chapter: Zusammenfassung und Ausblick (Summary and Outlook)	5
0.3 Scientific Writing Style	6
0.3.1 General Principles	6
0.3.2 Sentence Construction	6
0.3.3 Grammar	6
0.3.4 Technical Language	6
0.3.5 Expression and Phrasing	6
0.3.6 Content Guidelines	6
0.3.7 Introductions and Transitions Between Sections	7
0.3.8 Consistency and Formatting Rules	7
0.4 Citations and Bibliography	7
0.4.1 Citation Style	7
0.4.2 Citation Placement	7
0.4.3 Direct Quotes	7
1 Introduction	8
1.1 Motivation and Context	8
1.2 Problem Statement and Objectives	8
1.3 Thesis Outline	8
2 Background	9

2.1 Reinforcement Learning for Robotic Manipulation (2.1)	9
2.2 Physics Simulation in Isaac Sim 5.1 (2.2)	9
2.3 NVIDIA Isaac Sim and IsaacLab Architecture (2.3)	9
2.4 Tendon-Driven Robot Kinematics and Modeling (2.4)	9
3 State of the Art and Research Gap	10
3.1 GPU-Accelerated RL for Manipulation (3.1)	10
3.2 Tendon- and Cable-Driven Robot Simulation (3.2)	10
3.3 Workspace Analysis Methods (3.3)	10
3.4 Identified Research Gap (3.4)	10
3.5 Success Criteria (3.5)	10
4 Methodology	11
4.1 Overview of Approach (4.1)	11
4.2 Physical Robot Design (4.2)	11
4.2.1 Kinematic Chain and Degrees of Freedom (4.2.1)	11
4.2.2 Antiparallelogram Elbow Mechanism (4.2.2)	11
4.2.3 Cable-Driven Wrist (4.2.3)	11
4.2.4 Drive System and Tendon Routing (4.2.4)	11
4.3 Simulation Model Construction (4.3)	11
4.3.1 Mesh Processing Pipeline (4.3.1)	11
4.3.2 Articulation and Physics Configuration (4.3.2)	11
4.3.3 Disc Approximation Variant (4.3.3)	11
4.3.4 Physical Antiparallelogram Variant (4.3.4)	11
4.3.5 Five-DoF Assembly (4.3.5)	11
4.4 Tendon Actuation in Simulation (4.4)	11
4.4.1 Jacobian-Transpose Torque Mapping (4.4.1)	11
4.4.2 Physical Body-Force Tendon Model (4.4.2)	11
4.4.3 Actuation Mode Comparison (4.4.3)	11
4.4.4 Motor Specifications and Simulation Parameters (4.4.4)	11
4.5 Model Validation (4.5)	11
4.5.1 Step-Response Testing Protocol (4.5.1)	11
4.5.2 Monte Carlo Workspace Analysis (4.5.2)	11
4.5.3 PD Gain Tuning (4.5.3)	11
4.6 Reinforcement Learning Task Formulation (4.6)	11
4.6.1 Simulation Environment (4.6.1)	12
4.6.2 Reach Task (4.6.2)	12
4.6.3 Cube Place Task (4.6.3)	12
4.6.4 Cube Sort Task — Outlook (4.6.4)	12
4.7 Training Infrastructure (4.7)	12
4.7.1 PPO Configuration (4.7.1)	12
4.7.2 Simulation Parameters (4.7.2)	12
4.7.3 Software and Hardware (4.7.3)	12
5 Results	13
5.1 Model Validation Results (5.1)	13
5.2 Reach Task Results (5.2)	13
5.3 Cube Place Results (5.3)	13

5.4 Cube Sort Preliminary (5.4)	13
5.5 Summary of Key Results (5.5)	13
6 Discussion	14
6.1 Evaluation of Outcomes (6.1)	14
6.2 Comparison with Related Work (6.2)	14
6.3 Limitations of the Study (6.3)	14
6.4 Practical Implications (6.4)	14
6.5 Recommendations and Lessons Learned (6.5)	14
7 Conclusion	15
7.1 Conclusion (7.1)	15
7.2 Outlook (7.2)	15
8 Appendices	16
Background — Citation Verification Log	17
8.1 §2.1 Reinforcement Learning for Robotic Manipulation	17
8.2 §2.2 Tendon-Driven Mechanisms and Linkage Kinematics	19
8.3 §2.3 Workspace Analysis	22
8.4 §2.4 Physics Simulation	23
8.5 §2.5 NVIDIA Isaac Sim and IsaacLab	25
8.6 §2.6 Robot and Scene Description Formats	27
8.7 §2.6 Prior Work (research-gap framing)	29
8.8 Provenance & next-pass policy	33
Bibliography	34

Guideline

0.1 Format and Style

0.1.1 Page Layout and Typography

- Paper format: white DIN A4 paper
- Font: Arial, 12pt (in this Typst template, Helvetica achieves this)
- Line spacing: 1.2 (the template uses 1.25)
- Text alignment: justified (Blocksatz) with automatic hyphenation enabled
- Font emphasis: use only in exceptional cases; headings follow the template style and must not be underlined
- Bullet points: only square bullet points (FAPS green), never round ones; use the template's built-in list style
- FAPS color scheme: FAPS-Grün (RGB 151, 193, 57) and white

0.1.2 Page Numbering and Binding

- Page numbering starts with the introduction (Arabic numerals)
- Directories (table of contents, list of figures, etc.) receive separate Roman numeral numbering; the cover page is not counted
- Print single-sided unless otherwise agreed with the supervisor
- Two bound copies required: adhesive binding with clear cover foil and green (FAPS green) back sheet
- Each copy must include a CD/DVD containing:
 - Digital version of the thesis (PDF)
 - All digital sources (papers, scanned documents, websites)
 - All images, tables, measurement data, models, and other data

0.1.3 Section and Chapter Formatting

- Chapters are numbered numerically with at most three levels (e.g., 1.1.1); no trailing dot after the last digit
- If a subsection X.1 exists, there must also be at least an X.2 (applies to all levels)
- Main chapters should have at least 3 subchapters; aim for a balanced structure (e.g., 5 chapters at level 1, each with ~4 at level 2, each with ~3 at level 3)
- Headings in the table of contents must match the headings in the text verbatim
- Every heading must be followed by actual content (including introductory text at the start of main chapters)

0.1.4 Paragraphs, Spacing, and Whitespace

- Each paragraph must consist of at least two sentences
- No blank line between paragraphs within a chapter; one blank line before the next heading at the end of a chapter
- No double spaces anywhere in the document; use search-and-replace to eliminate them before submission
- Use non-breaking spaces () between:

- Titles and names: “Prof. Dr. Beispielhausen”
- Date expressions: “19. September 2007”
- Abbreviations: “z. B.”, “u. a.”, “d. h.”
- Numbers and units: “44 mm”
- Numbers and words: “1.350 Personen”, “No 5”
- Citation references
- Check automatic hyphenation carefully for errors

0.1.5 Figures

- Every figure receives a caption below it (Unterschrift), formulated so the figure is understandable without the surrounding text
- All figures are numbered consecutively with Arabic numerals and listed in the list of figures
- Every figure must be referenced in the body text
- Title image: no caption, no numbering, not included in the list of figures
- Appendix figures: numbered within each appendix separately; not included in the list of figures
- Ensure sufficient image quality; recreate figures if originals are blurry, low-resolution, or scanned
- Use the same font in figures as in body text (Arial); font size 10pt is acceptable but must be consistent across all figures
- All figure components must be fully labeled: photo components, axes, parameters, legends with complete process parameters
- Fill areas with color only if the color conveys meaningful information; otherwise keep figures clean and minimal
- For recreated/modified figures from sources, cite with “in Anlehnung an [xy]” (based on [xy])
- Use a consistent color scheme across all figures; preferably FAPS colors unless industry cooperation requires otherwise
- Use a self-created graphic on the first page of the thesis

0.1.6 Tables

- Every table receives a caption above it (Überschrift) with a meaningful, descriptive title
- Tables are numbered consecutively with Arabic numerals and listed in the table directory
- Every table must be referenced in the body text
- Use the same font as in body text; maintain consistent font size across all tables
- Preferred alignment inside tables: left-aligned or centered (not justified)
- If a table spans a page break, do not break within a row; repeat the table header on the new page
- Extensive tables belong in the appendix

0.1.7 Diagrams

- Follow DIN 461 for axis labeling in diagrams
- Label axes with an arrow in the direction of increase (except for trial numbers etc.)
- Replace the axis label at the second-to-last tick mark with the unit
- Display all relevant information within the diagram
- Place a frame around diagrams
- Use Arial font inside diagrams with a legible font size
- Diagrams are captioned and numbered as figures (no separate numbering type)

0.1.8 Formulas

- Formula numbers are enclosed in round parentheses
- Numbering is chapter-based in the form X.Y (X = chapter number, Y = formula number within the chapter)
- Numbering restarts within each main chapter
- Variables used in a formula must be explained directly below the formula in tabular form

0.1.9 Appendix

- Technical drawings, extensive tables, supplementary documents, etc. go into appendices
- Each appendix starts on a new page and may span multiple pages
- Appendices are numbered with consecutive uppercase Latin letters (Appendix A, Appendix B, etc.)

0.1.10 Eidesstattliche Erklärung (Statutory Declaration)

- A signed declaration of independent authorship must be included on a separate page without a page number, placed after the cover page
- It must be signed by hand

0.2 Thesis Structure and Content

0.2.1 Required Document Order

1. Deckblatt (Cover Page)
2. Erklärung des Kandidaten (Statutory Declaration)
3. Inhaltsverzeichnis (Table of Contents)
4. Abbildungsverzeichnis (List of Figures)
5. Tabellenverzeichnis (List of Tables)
6. Abkürzungsverzeichnis (List of Abbreviations)
7. Einleitung und Zielstellung (Introduction and Objective)
8. Grundlagen / Stand der Technik (Background / State of the Art)
9. Handlungsbedarf (Research Gap / Need for Action)
10. Methodenteil (Methodology)
11. Ergebnisteil (Results)
12. Diskussionsteil (Discussion)
13. Zusammenfassung und Ausblick (Summary and Outlook)
14. Literaturverzeichnis (Bibliography)
15. Anhang (Appendix)
16. Lebenslauf (Curriculum Vitae)

0.2.2 Page Count Guidelines

- Bachelorarbeit: ca. 60–80 pages
- Projekt-/Studienarbeit: ca. 60–80 pages
- Master-/Diplomarbeit: ca. 80–120 pages
- These counts exclude appendices and directories

0.2.3 Red Thread (Roter Faden)

- A clear red thread must run through the entire thesis, starting from the central research question in the introduction
- The reader must always be able to see how any part of the thesis relates to the research question
- The thesis must proceed purposefully and not lose itself in tangential topics
- All claims and statements must be causally justified; results must not merely be listed but convincingly explained
- Each main chapter must begin with a short introduction motivating the content and end with a transition to the next
- Verify the red thread by checking all headings alone (without reading the text)—the logical flow should be recognizable

0.2.4 Chapter: Einleitung und Zielstellung (Introduction)

- Length: approximately 2 pages, maximum 3 pages
- Purpose: establish contact with the reader; provide orientation (“What can the reader expect?”)
- Content requirements:
 - Introduce the topic and its general background; explain why it is relevant
 - Present the specific research question
 - Justify the scope and boundaries of the thesis (what is included, what is excluded, and why)
 - Optionally describe the state of research (to show what exists and where gaps are)
 - Explain the methodological approach
 - Outline the structure of the thesis for the reader

0.2.5 Chapter: Grundlagen und Stand der Technik (Background and State of the Art)

- Clarify central concepts and define relevant terms
- Present the current state of the art objectively and without judgment (evaluation happens in the next chapter)
- Only cover theories, definitions, and knowledge directly relevant to the research question; avoid overly broad coverage
- If multiple competing theories exist, provide an overview
- Definitions are only needed for terms not commonly known by domain experts
- Present content in your own words; near-verbatim copying or translation from sources counts as insufficient
- Use primarily English-language sources where possible
- Prefer journal articles and conference papers over websites
- Avoid popular-science sources; only quote newspapers for current events
- Internet sources are acceptable only if no current literature exists elsewhere (e.g., software documentation); verify their credibility

0.2.6 Chapter: Handlungsbedarf (Research Gap)

- Critically discuss the literature from the previous chapter regarding unsolved problems and errors
- Derive current deficits and research gaps; formulate the specific research question

- Explain who considers the problem significant and what its broader impact is
- Describe the contribution the thesis makes to both research (theory, models, methods, facts) and practice
- Narrow the problem statement into a concrete, achievable objective with clear expected outcomes

0.2.7 Chapter: Methodenteil (Methodology)

- Together with Results and Discussion, this forms the main body (~2/3 of the thesis)
- Describe the investigation plan, procedures, and all steps and tools in sufficient detail for reproducibility
- Justify why specific methods were chosen over alternatives (pros/cons analysis)
- For experimental studies: describe the population and sample in all relevant characteristics; justify sample adequacy
- Scientific criteria for methods: objectivity, reliability, and validity
- The description must enable other researchers to reproduce the investigation
- Only describe analysis procedures in detail if they are novel or non-standard

0.2.8 Chapter: Ergebnisteil (Results)

- Present results of the investigation objectively and without interpretation
- Structure the chapter logically, aligned with the theory and methodology sections
- Use figures, tables, and diagrams to visualize results
- No interpretation or evaluation at this stage

0.2.9 Chapter: Diskussionsteil (Discussion)

- Summarize and critically discuss the results
- Evaluate and interpret results in relation to the original problem and objective from the introduction
- Argue the significance for scientific discourse and practical implications

0.2.10 Chapter: Zusammenfassung und Ausblick (Summary and Outlook)

- Maximum 3 pages
- Reconnect to the central problem and objective from the introduction
- Compactly summarize the most important results
- Place results in a broader context
- Address:
 - What are the key findings with respect to the research questions?
 - How can the results be interpreted or evaluated?
 - What questions remain open, and why?
 - What do the results mean for future work on this topic?
 - Where are opportunities for further research?
- Discuss the limitations of the investigation
- Provide suggestions for future work and improvements

0.3 Scientific Writing Style

0.3.1 General Principles

- Write in present tense (Präsens) and nominal style (Nominalstil)
- Use impersonal form throughout; never use “ich” (I) or “man” (one/you)
- Develop the red thread of argumentation before formulating sentences
- Every sentence and every claim must be precise and well-placed
- Write result-oriented: “Die Einflüsse führen zu Veränderungen.” not “Im Rahmen der Arbeit wurden untersucht...”

0.3.2 Sentence Construction

- Use passive voice sparingly—prefer active formulations
- Avoid sentences with “man” entirely
- Be careful with infinitive constructions
- Avoid chaining too many nouns together (“Substantivaneinanderreihungsproblem”)
- No nested or overly complex sentences (Schachtelsätze)

0.3.3 Grammar

- Avoid relative time references (“kürzlich”, “später”, “modern”) since the thesis should remain valid for years
- Do not use subjunctive mood (könnte, sollte, müsste, dürfte)

0.3.4 Technical Language

- Use precise adjectives; avoid vague terms like “innovativ” or “intelligent”
- No pleonasms (redundant word pairs)
- Do not artificially inflate terms
- Be cautious with superlatives and comparative forms
- Do not invent words
- Use anglicisms and foreign words sparingly; only when no equivalent German term exists
- Use precise technical terminology; avoid colloquial language
- Use technical terms consistently throughout

0.3.5 Expression and Phrasing

- Spell out numbers from zero to twelve; use digits from 13 onward
- Avoid colloquialisms and journalistic clichés
- Do not use rhetorical questions; make clear statements instead
- Avoid vague qualifiers: “im Allgemeinen”, “in der Regel”, “beispielsweise”, “teilweise”
- Avoid filler words: “nämlich”, “also”, “so”, “eben”
- Avoid repetition of both words and arguments
- Always formulate concretely and precisely; define unambiguously; enumerate completely; avoid generalizations
- Use nominal style

0.3.6 Content Guidelines

- Do not document iterative development processes or suboptimal intermediate solutions; present only the final approach

- Do not describe device/hardware development in excessive detail; focus on scientifically relevant, generalizable, transferable aspects
- Use direct quotes rarely and only pointedly; prefer evaluating and developing ideas further in your own words
- Introduce abbreviations only when necessary for space; spell out on first use and include in the abbreviation index
- Emphasize scientific quality rather than engineering challenges
- Emphasize fundamental, generalizable, and transferable findings rather than application-specific details
- Instead of giving examples, enumerate completely, structure, and evaluate
- Make all enumerations and representations as complete and non-overlapping as possible (MECE principle)
- Always support claims with your own analyses or cited sources
- No implied personal opinion and no subjective assessments except in the Outlook section
- Where possible, summarize each chapter's key message in a figure for quick reference

0.3.7 Introductions and Transitions Between Sections

- Every main chapter or main section must begin with a brief introduction and end with an outlook/transition to the next section
- When introducing detailed explanations, name all relevant points in the introductory sentence

0.3.8 Consistency and Formatting Rules

- Introduce each abbreviation exactly once and then use it consistently
- Never write a paragraph consisting of only one sentence (minimum two sentences)
- Never write a list with only one bullet point (minimum two)
- Use nominal style for bullet-point items
- Maintain consistent spelling of expressions throughout

0.4 Citations and Bibliography

0.4.1 Citation Style

- Citation style: IEEE (adapted FAPS style)
- Reference sources in the text using numbers in square brackets: [xy]
- Multiple consecutive citations use separate brackets
- The bibliography is ordered chronologically and numbered consecutively
- Only actually cited works appear in the bibliography

0.4.2 Citation Placement

- A citation before the period refers to the specific sentence.
- A citation after the period of the last sentence refers to the entire paragraph.

0.4.3 Direct Quotes

- Must be letter-perfect and character-perfect reproductions
- Enclose in quotation marks; cite the source at the end

1 Introduction

1.1 Motivation and Context

- Tendon-driven robots: compliant design, reduced distal inertia, complex actuation modeling
- Reinforcement learning as a data-driven alternative to analytical control
- GPU-accelerated simulation (Isaac Sim + IsaacLab) as the enabling platform
- Frame the PA as the simulation foundation of a larger textile-sorting project at FAPS

1.2 Problem Statement and Objectives

- **Objective 1:** Validated tendon-driven robot model in Isaac Sim
- **Objective 2:** Progressive RL-based manipulation (reach then cube place), comparing PD vs tendon actuation
- **Objective 3:** Documented infrastructure foundation for the master thesis
- Scope exclusion: no cloth, no sorting, no sim-to-real, no wrist transfer (deferred to MA)

1.3 Thesis Outline

- Brief chapter summary for reader orientation

2 Background

2.1 Reinforcement Learning for Robotic Manipulation (2.1)

2.2 Physics Simulation in Isaac Sim 5.1 (2.2)

2.3 NVIDIA Isaac Sim and IsaacLab Architecture (2.3)

2.4 Tendon-Driven Robot Kinematics and Modeling (2.4)

3 State of the Art and Research Gap

3.1 GPU-Accelerated RL for Manipulation (3.1)

3.2 Tendon- and Cable-Driven Robot Simulation (3.2)

3.3 Workspace Analysis Methods (3.3)

3.4 Identified Research Gap (3.4)

3.5 Success Criteria (3.5)

4 Methodology

4.1 Overview of Approach (4.1)

4.2 Physical Robot Design (4.2)

4.2.1 Kinematic Chain and Degrees of Freedom (4.2.1)

4.2.2 Antiparallelogram Elbow Mechanism (4.2.2)

4.2.3 Cable-Driven Wrist (4.2.3)

4.2.4 Drive System and Tendon Routing (4.2.4)

4.3 Simulation Model Construction (4.3)

4.3.1 Mesh Processing Pipeline (4.3.1)

4.3.2 Articulation and Physics Configuration (4.3.2)

4.3.3 Disc Approximation Variant (4.3.3)

4.3.4 Physical Antiparallelogram Variant (4.3.4)

4.3.5 Five-DoF Assembly (4.3.5)

4.4 Tendon Actuation in Simulation (4.4)

4.4.1 Jacobian-Transpose Torque Mapping (4.4.1)

4.4.2 Physical Body-Force Tendon Model (4.4.2)

4.4.3 Actuation Mode Comparison (4.4.3)

4.4.4 Motor Specifications and Simulation Parameters (4.4.4)

4.5 Model Validation (4.5)

4.5.1 Step-Response Testing Protocol (4.5.1)

4.5.2 Monte Carlo Workspace Analysis (4.5.2)

4.5.3 PD Gain Tuning (4.5.3)

4.6 Reinforcement Learning Task Formulation (4.6)

4.6.1 Simulation Environment (4.6.1)

4.6.2 Reach Task (4.6.2)

4.6.3 Cube Place Task (4.6.3)

4.6.4 Cube Sort Task — Outlook (4.6.4)

4.7 Training Infrastructure (4.7)

4.7.1 PPO Configuration (4.7.1)

4.7.2 Simulation Parameters (4.7.2)

4.7.3 Software and Hardware (4.7.3)

5 Results

5.1 Model Validation Results (5.1)

5.2 Reach Task Results (5.2)

5.3 Cube Place Results (5.3)

5.4 Cube Sort Preliminary (5.4)

5.5 Summary of Key Results (5.5)

6 Discussion

6.1 Evaluation of Outcomes (6.1)

6.2 Comparison with Related Work (6.2)

6.3 Limitations of the Study (6.3)

6.4 Practical Implications (6.4)

6.5 Recommendations and Lessons Learned (6.5)

7 Conclusion

7.1 Conclusion (7.1)

7.2 Outlook (7.2)

8 Appendices

- Appendix A: Simulation and Control Parameters
- Appendix B: PPO Hyperparameters
- Appendix C: Cube Place Reward Terms

Background — Citation Verification Log

This appendix records, per Background subsection, which page / section / quotation in each cited PDF supports the corresponding claim in the thesis. Each bullet shows the citation key, a brief evidence pointer, and a (PDF) link to the local source file under `doc/Literatur/sources/`. Line references in square brackets (e.g. [2_2:18]) point at the line in the corresponding chapters/background/2_*.typ file where the claim appears.

8.1 §2.1 Reinforcement Learning for Robotic Manipulation

- **V1** — [1], [2] [2_1:9] — OK: textbook + survey introduced as canonical references for RL formalism and robotics-RL constraints respectively; Sutton & Barto Ch. 3 “Finite Markov Decision Processes” p. 47, Kober2013 §2.1 pp. 4–5 (“Reinforcement Learning in the Context of Classical Robotics”). [SuttonBarto2018 PDF](#) — [Kober2013 PDF](#)
- **V2** — [1] [2_1:14] — OK: MDP tuple $(\mathcal{S}, \mathcal{A}, P, r, \gamma)$ matches Ch. 3 §3.1 p. 47 (“Finite Markov Decision Processes”); discounted return $G_t = \sum \gamma^k r_{t+k+1}$ p. 76 §3.4; performance measure $J(\theta) = \mathbb{E}_{\pi_\theta}[\sum \gamma^t r_t]$ matches policy-gradient setup §13.2 p. 324 (“we seek to maximize performance, so its updates approximate gradient ascent in J”). [PDF](#)
- **V3** — [3], [1], [2] [2_1:16] — OK (resolved 2026-05-02): partial-observability sentence now cites a formal POMDP reference (Spaan2012, *Reinforcement Learning: State-of-the-Art* Springer 2012, Ch. 12 “Partially Observable Markov Decision Processes” — defines the POMDP tuple $(\mathcal{S}, \mathcal{A}, T, R, \Omega, O)$ with observation model $O(o | s, a)$ and history-dependent optimal policy) plus Sutton & Barto §17.3 “Partially Observable States” p. 465; the practical recurrent-policy / observation-stacking claim is supported by Kober2013 §2.1 pp. 4–5. [Spaan2012 PDF](#) — [SuttonBarto2018 PDF](#) — [Kober2013 PDF](#)
- **V4** — [2] [2_1:23-29] — OK: the three “key challenges” enumeration (continuous high-DoF control, contact-rich nonlinear dynamics, sparse/delayed feedback) maps to §1 p. 2 (“dimensionality, real-world samples, under-modeling and uncertainty, goal specification”) and §3 “Challenges in Robot Reinforcement Learning”. [PDF](#)
- **V5** — [2], [1] [2_1:32] — OK: dense-vs-sparse reward / shaping discussion in Kober §6 “Tractability Through Reward” p. 17 (“the choice of an appropriate reward function ... is often a delicate matter”); Sutton & Barto §17.4 “Designing Reward Signals” p. 469 mirrors the trade-off. [Kober2013 PDF](#) — [SuttonBarto2018 PDF](#)
- **V6** — [4] [2_1:35] — OK (resolved 2026-05-02): the equation $F(s, a, s') = \gamma \Phi(s') - \Phi(s)$ matches Theorem 1 p. 4 of Ng, Harada, Russell 1999 (ICML). Local PDF replaced with the Stuart Russell faculty-page copy (Berkeley, MD5 9e20516d...); page-4 text now extractable via PyMuPDF. Verbatim Theorem 1 statement: “that F is a potential-based shaping function is a necessary and sufficient condition for it to guarantee consistency with the optimal policy” (sufficiency + necessity sub-clauses follow). [PDF](#)
- **V7** — [5] [2_1:37] — OK: AAMAS 2012 paper title “Dynamic Potential-Based Reward Shaping” abstract p. 1: dynamic (time-varying) potential functions preserve single-agent optimal policy invariance and Nash equilibria — directly supports the “even when the potential changes during an episode” claim. [PDF](#)
- **V8** — [6] [2_1:39] — OK: Lego brick-stacking task introduced p. 1 right column; §VI “Composite Shaping Rewards” p. 6 introduces phase-gated reward components; sparse

reward subtask list p. 7 (“STACK”, “STACK_AND_LEAVE”) matches the gated-phase pattern cited. [PDF](#)

- **V9** — [7] [2_1:41] — OK: JAIR 73 (2022) paper “Reward Machines: Exploiting Reward Function Structure in Reinforcement Learning” — abstract and §1 p. 2 define finite-state reward machines as automata with per-state reward functions driven by propositional events; surname **Toro Icarte** (preserved correctly in thesis prose). [PDF](#)
- **V10** — [8], [9] [2_1:43] — OK (resolved 2026-05-02): the robosuite paper (arXiv:2009.12293) supports the conceptual phase-gated reward claim via §3.2 “Modular Design”; the literal API name `staged_rewards()` originates in the source file `robosuite/environments/manipulation/pick_place.py` (method definition with the docstring “Returns staged rewards based on current physical states. Stages consist of reaching, grasping, lifting, and hovering”; reach/grasp/lift/hover multipliers 0.1/0.35/0.5/0.7). A local HTML snapshot of the GitHub blob page (commit `aaa8b9b`, accessed 2026-05-02) is filed alongside the bib entry `@robosuite_pickplace_repo`. [Zhu2020 PDF](#) — [robosuite pick_place.py snapshot](#)
- **V11** — [1] [2_1:47] — OK: actor–critic methods Ch. 13.5 p. 331; advantage function $A^{\pi(s,a)} = Q^{\pi(s,a)} - V^{\pi(s)}$ §13.4 p. 329; baseline-in-policy-gradient discussion §13.4. [PDF](#)
- **V12** — [10] [2_1:50] — OK: ICLR 2016 preprint, abstract p. 1 “an exponentially-weighted estimator of the advantage function that is analogous to $TD(\lambda)$ ”; bias–variance trade-off discussed §3 p. 4; GAE definition Eq. (15) on p. 5 $\hat{A}_t^{\text{GAE}(\gamma,\lambda)} = \sum (\gamma\lambda)^l \delta_{t+l}^V$. [PDF](#)
- **V13** — [1], [2] [2_1:54] — OK: exploration–exploitation trade-off Sutton & Barto §1.1 p. 3 (“the exploration–exploitation dilemma”) and Ch. 2; Kober survey §3.1 “Curse of Dimensionality” p. 6 discusses long-horizon and unsafe-state exploration in robot-RL. [SuttonBarto2018 PDF](#) — [Kober2013 PDF](#)
- **V14** — [1], [11], [8] [2_1:56] — OK: stochastic policies Sutton & Barto §13.1 p. 322; Kalashnikov2018 (QT-Opt) p. 5 §IV explicitly defines the action vector $a \in \mathcal{A}$ as Cartesian translation $t \in \mathbb{R}^3$, sin/cos rotation $r \in \mathbb{R}^2$, **binary gripper open/close commands** $g_{\text{open}}, g_{\text{close}}$, plus a termination flag — exactly the hybrid-action-space cited. Zhu2020robosuite §3 documents the parallel-jaw gripper action interface. [SuttonBarto2018 PDF](#) — [Kalashnikov2018 PDF](#) — [Zhu2020 PDF](#)
- **V15** — [2] [2_1:58–59] — OK: curriculum learning / progressive task difficulty Kober survey §6.1 “Reward Shaping” p. 17 and §5 “Reinforcement Learning in the Real World” pp. 14–15; episode-reset / state-distribution discussion §3.2 p. 7 (“most reinforcement-learning algorithms require ... resetting”). [PDF](#)
- **V16** — [2], [1] [2_1:61] — OK: closing-paragraph reference to environment-design dependence reuses prior verifications V13/V15. [Kober2013 PDF](#) — [SuttonBarto2018 PDF](#)
- **V17** — [2], [12], [13] [2_1:65] — OK: simulation-training-rationale Kober survey §5 p. 14 (“simulations ... cheap to gather and inherently safe”); Tobin2017 abstract p. 1 “we randomize the renderer to expose the model to a wide range of environments at training time”; Peng2018 §IV.C p. 4 “Dynamics Randomization” lists mass, damping, motor gains, latency. [Kober2013 PDF](#) — [Tobin2017 PDF](#) — [Peng2018 PDF](#)
- **V18** — [10], [14], [2] [2_1:69–73, eq:ppo_clip] — OK: PPO clipped surrogate equation transcription verified verbatim against Schulman2017 Eq. (7) p. 3: “ $L^{\text{CLIP}(\theta)} = \mathbb{E}_t[\min(r_{t(\theta)}\hat{A}_t, \text{clip}(r_{t(\theta)}, 1-\epsilon, 1+\epsilon)\hat{A}_t)]$ ”; importance ratio $r_{t(\theta)} = \frac{\pi_{\theta(a_t|s_t)}}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$ defined directly above Eq. (6) on the same page; pairing with GAE for $\hat{A}_t$ stated §6.1 p. 6. Kober2013 §6 supports the contact-rich-stability rationale. [Schulman2015GAE PDF](#) — [Schulman2017PPO PDF](#) — [Kober2013 PDF](#)

- **V19** — **[14]** [2_1:79] — OK: stable-update claim restates Eq. (7) clipping motivation p. 3 (“the second term ... removes the incentive for moving r_t outside of the interval $[1 - \varepsilon, 1 + \varepsilon]$ ”). [PDF](#)
- **V20** — **[1]** [2_1:80] — OK: Gaussian policies for continuous actions Sutton & Barto §13.7 p. 335 (“Policy Parameterization for Continuous Actions”). [PDF](#)
- **V21** — **[2]** [2_1:81] — OK: rollout-and-update training-loop fit for robotics Kober §4.2 “Policy Search” p. 12 and §5 “Reinforcement Learning in the Real World” p. 14. [PDF](#)

8.2 §2.2 Tendon-Driven Mechanisms and Linkage Kinematics

- **V1** — **[15], [16]** [2_2:13] — OK: Rao2021 (Front. Robot. AI 8:630245, 2021) “How to Model Tendon-Driven Continuum Robots and Benchmark Modelling Performance” — covers TDCR principles, accepted boundary use for general tendon-actuation framing. Nemoto2022 (IEEE RA-L 7(2):5727, 2022) abstract p. 1: “highly dynamic 2-DOF cable-driven robotic wrist based on a novel topology” — directly supports remote-actuation / low-distal-inertia claim. [Rao2021 PDF](#) — [Nemoto2022 PDF](#)
- **V2** — **[17], [18], [19]** [2_2:16] — OK: Kobayashi1998 abstract p. 1: “issues are discussed regarding tendon redundancy and joint stiffness adjustability for a robotic mechanism driven with redundant tendons”; §3 defines driving redundancy $\alpha = M - N$, §4 derives bias tension $J^T f_b = 0$ for null-space co-contraction. Muralidharan2021 p. 1 “Optimal design and comparative study of two antagonistically actuated tensegrity joints” — direct support for antagonistic-pair / stiffness-modulation. TsaiLee1989 (ASME J. Mech. Trans. Auto. Design 111(1):59–65, scanned PDF) introduces the $n+1$ -cable rule for force closure in graph-theoretic tendon-driven mechanisms (PDF text-layer is image-only; metadata-confirmed authorship Tsai & Lee). The $n+1$ result is also independently attested by Pott2009 (V14 below) and Kobayashi1998 §3. [Kobayashi1998 PDF](#) — [Muralidharan2021 PDF](#) — [TsaiLee1989 PDF](#)
- **V3** — **[20], [21]** [2_2:31] — OK: Miyasaka2020 (IEEE/ASME T-Mech 25(2):1095, 2020) p. 1 title+abstract: “Modeling Cable-Driven Robot With Hysteresis and Cable–Pulley Network Friction” — direct match for cable stretch + friction + hysteresis. Kato2016 (Int. J. CARS, 2016) p. 1 title: “Tendon-driven continuum robot for neuroendoscopy: validation of extended kinematic mapping for hysteresis operation” — supports model-based hysteresis compensation. [Miyasaka2020 PDF](#) — [Kato2016 PDF](#)
- **V4** — **[22], [23]** [2_2:38] — OK: McCarthy2011 (“Geometric Design of Linkages”, Springer 2nd ed., 2011, doi 10.1007/978-1-4419-7892-9) is the canonical four-bar / loop-closure reference. Uicker2017 (“Theory of Machines and Mechanisms”, OUP 5th ed., 2017) §1.9 “Grashof’s Law” p. 33 (TOC entry verified, p. 33 in book / p. 13 of PDF) — supports the Grashof classification claim. [McCarthy2011 PDF](#) — [Uicker2017 PDF](#)
- **V5** — **[22], [23]** [2_2:40-42, eq:bg_loop_closure] — OK (resolved 2026-05-02): co-cited Uicker2017 alongside McCarthy2011 to anchor the complex-exponential form $Z_1 e^{i\theta} + Z_2 e^{i\beta} = Z_3 e^{i\varphi} + Z_4$. Uicker/Sandor/Erdman is the canonical home of the complex-number loop-closure (Uicker2017 Ch. 2 “Position, Posture, and Displacement” / §2.10 “Complex-Number Methods”), while McCarthy2011 covers loop closure in vector/screw form. The equation as written is correct under either textbook’s notation. [McCarthy2011 PDF](#) — [Uicker2017 PDF](#)

- **V6** — [23], [24] [2_2:42] — OK: Uicker2017 (transmission-angle classical treatment, §3 Mechanical Advantage / §4 Velocity Analysis). Hartenberg1955 (Hartenberg & Denavit, J. Appl. Mech. 22:215, 1955; the foundational kinematic-notation paper) — supports the transmission-angle / near-singular-configuration claim as a historical primary source. [Uicker2017 PDF](#) — [Hartenberg1955 PDF](#)
- **V7** — [25], [22] [2_2:50] — OK (with PDF text-layer caveat): Dijksman1977 (MMT 12(1), 1977) is the canonical antiparallelogram-topology reference; the local PDF's first 12 pages do not yield the keyword `centrode` via PyMuPDF (likely scanned/font-subset), but the paper's title and Stachel2000's explicit citation of Dijksman as the centrode source (V8) confirm the attribution. McCarthy2011 covers crossed four-bar variants in the four-bar taxonomy chapter. [Dijksman1977 PDF](#) — [McCarthy2011 PDF](#)
- **V8** — [25], [26] [2_2:53] — OK (Bryant2013 dropped 2026-04-24): Stachel2000 p. 3 verbatim: **“under the reflection in any tangent line l of the ellipse e (= fixed polode) with focal points a'_1, b'_1 and semi-axes β and $\sqrt{\beta^2 - \alpha^2}$ ”** — exact match for the cited semi-axes formula. Dijksman1977 attributed by Stachel as the original ellipse-centrode result. The earlier Bryant2013 algebraic-rigidity reference was removed from the chapter (the local PDF was mislabeled and the claim is non-essential). [Stachel2000 PDF](#) — [Dijksman1977 PDF](#)
- **V9** — [27] [2_2:58] — OK: Furet & Wenger 2019 (ASME J. Mechanisms & Robotics 11(6):060904) p. 1 title: “Kinetostatic Analysis and Actuation Strategy of a Planar Tensegrity 2-X Manipulator” — kinetostatic analysis explicitly addresses the variable-lever-arm / configuration-dependent transmission of the X-joint. [PDF](#)
- **V10** — [27] [2_2:61] — OK: parallelogram/antiparallelogram bifurcation handling at collinear singularity is part of the kinetostatic analysis in Furet2019 (cable-routing prevents mode-switching by maintaining the crossed configuration). [PDF](#)
- **V11** — [28] [2_2:71] — OK: Pott2018 (“Cable-Driven Parallel Robots — Theory and Application”, Springer Tracts in Advanced Robotics 120, 2018) — canonical CDPR textbook; Ch. 2 introduces the cables-only-tension constraint and platform-with- m -cables model. [PDF](#)
- **V12** — [28], [29] [2_2:72] — OK: Pott2018 Ch. 2 (cables-only-tension) and Verhoeven2004 (TU Duisburg PhD thesis “Analysis of the Workspace of Tendon-based Stewart Platforms”) cover the tensile-only constraint. [Pott2018 PDF](#) — [Verhoeven2004 PDF](#)
- **V13** — [29], [30] [2_2:75-79, eq:bg_structure_matrix] — OK: equation $w = A^T t$ verbatim verified — Verhoeven2004 p. 10 nomenclature: **“ A^T structure matrix Eq. (2.24), p. 26”** — defines the structure-matrix notation cited in the thesis. Bruckmann2008a p. 3 confirms the wire-robot/Stewart-Gough-platform formalism. [Verhoeven2004 PDF](#) — [Bruckmann2008 PDF](#)
- **V14** — [30], [31], [29], [32], [33], [34] [2_2:79] — OK: structure-matrix-as-Jacobian-transpose / kineto-static-duality claim. Verhoeven attribution for “structure matrix” terminology (V13). Gouttefarde2006 uses equivalent “wrench matrix” (V15). Craig2005 Ch. 5 “Jacobians: velocities and static forces” (TOC verified, p. 135) and Murray1994 are the standard textbook references for the kineto-static duality. [Bruckmann2008 PDF](#) — [Pott2009 PDF](#) — [Verhoeven2004 PDF](#) — [Gouttefarde2006 PDF](#) — [Craig2005 PDF](#) — [Murray1994 PDF](#)
- **V15** — [32], [35], [31] [2_2:86] — OK: Gouttefarde2006 (IEEE TRO 22(3):434, June 2006) p. 1 title: “Analysis of the Wrench-Closure Workspace of Planar Parallel Cable-Driven Mechanisms” — direct definition of wrench-closure workspace + $n+1$ -cable requirement. Bosscher2006 (IEEE TRO 22(5):890, Oct 2006) p. 1: “Wrench-Feasible Workspace Generation for Cable-Driven Robots” — exact match for wrench-feasible-workspace concept with

bounded tensions. Pott2009 p. 1: “algorithm to determine feasible force distributions for parallel wire robots in closed form” — supports the closed-form $t = t_{\text{particular}} + N\lambda$ decomposition. [Gouttefarde2006 PDF](#) — [Bosscher2006 PDF](#) — [Pott2009 PDF](#)

- **V16** — **[34]** [2_2:102, transmission Jacobian eq] — OK: Murray1994 (“A Mathematical Introduction to Robotic Manipulation”, CRC, 1994) Ch. 3 “Manipulator Kinematics” + Ch. 4 “Robot Dynamics and Control” cover the differential-kinematics / transmission-Jacobian formalism $\dot{l} = R(q)\dot{q}$. [PDF](#)
- **V17** — **[33], [34], [36]** [2_2:105, eq:bg_jacobian_transpose] — OK: equation $\tau = R^T(q)T$ is the textbook-standard virtual-work derivation. Craig2005 Ch. 5 “Jacobians: velocities and static forces” p. 135 (TOC verified). Tsai1999 (“Robot Analysis: The Mechanics of Serial and Parallel Manipulators”, Wiley, 1999) covers the same derivation. [Craig2005 PDF](#) — [Murray1994 PDF](#) — [Tsai1999 PDF](#)
- **V18** — **[37], [19]** [2_2:108] — OK: Salisbury1982 (Stanford-JPL hand) p. 5 confirms Jacobian-rank/singularity discussion: “**Singularities occur where the rank of the Jacobian matrix becomes less than full**”; the transpose-mapping passage is on p. 6 lines 13–17 (cf. V21). TsaiLee1989 generalisation to graph-theoretic tendon-driven mechanisms now available locally as scanned PDF (metadata-confirmed authorship; cf. V2). [Salisbury1982 PDF](#) — [TsaiLee1989 PDF](#)
- **V19** — **[19], [27], [34]** [2_2:112] — OK: configuration-dependent vs constant Jacobian — MurrayLi1994 covers the general $R(q)$ formulation. Furet2019 confirms the X-joint produces a configuration-dependent transmission (V9). The TsaiLee1989 attribution for “circular pulleys yield constant R ” is supported by the now-available scanned PDF (cf. V2). [TsaiLee1989 PDF](#) — [Murray1994 PDF](#) — [Furet2019 PDF](#)
- **V20** — **[37], [19], [17]** [2_2:117] — OK: positive-tension constraint + $n+1$ rule (Salisbury1982 p. 5–6) and null-space co-contraction (Kobayashi1998 §4 — verified in V2). [Salisbury1982 PDF](#) — [Kobayashi1998 PDF](#)
- **V21** — **[38]** [2_2:130] — OK (Sultan2009 / Skelton2009 substituted 2026-04-24): the tensegrity definition + class- k taxonomy + Snelson–Fuller history is now cited via Shah2022 instead of Sultan2009 / SkeltonDeOliveira2009 (those PDFs are paywalled; Shah2022 covers all the same canonical claims). Shah2022 p. 2 verbatim: “**class 1 tensegrities such as the six-bar icosahedron ... have only one rigid body contacting each node**”; “Class 4 ‘tbar’ tensegrities (resembling the kites built by Snelson, but pinned at the two struts’ intersection)”. [Shah2022 PDF](#)
- **V22** — **[39], [40]** [2_2:134] — OK: Bicchi2004 p. 1: “**A robot arm that is to interact with humans has a single design consideration at a premium—safety**” (passive-compliance-for-safety claim, faster than control). Walter2023 (IFTToMM proceedings, paper title “Tensegrity-Inspired Joint Can Protect from Impacts by Isolating”, begins PDF p. 348 / chapter p. 344) p. 354 / chapter p. 350 conclusion: “**isolates impact forces up to 83 %**”. [Bicchi2004 PDF](#) — [Walter2023 PDF](#)
- **V23** — **[38]** [2_2:135] — OK (SkeltonDeOliveira2009 substituted 2026-04-24): high strength-to-weight / axial-only-loading claim now cited via Shah2022 (no local Skelton PDF available). Shah2022 p. 3 verbatim supports the axial-loading framing: “**each member in a tensegrity primarily only experiences tension or compression**”. [Shah2022 PDF](#)
- **V24** — **[18], [17]** [2_2:136] — OK: cable-pretension stiffness modulation — Muralidharan2021 p. 1 and Kobayashi1998 §4 (V2). [Muralidharan2021 PDF](#) — [Kobayashi1998 PDF](#)

- **V25** — [27], [41] [2_2:137] — OK: remote-actuation / minimal-distal-mass claim. Furet2019 (V9). Fasquelle2020 p. 1 title+abstract: “bio-inspired 3-DOF light-weight manipulator with tensegrity X-joints ... safe interactions” — direct match. [Furet2019 PDF](#) — [Fasquelle2020 PDF](#)
- **V26** — [38] [2_2:145] — OK: Shah2022 (Soft Robotics, 2022) abstract p. 1 verbatim: **“Tensegrity robots contain compliant networks of rigid struts and soft cables, allowing them to change their shape by adjusting their internal tension”** — direct match for rigid–soft bridge claim and locomotion-spectrum framing. [PDF](#)
- **V27** — [42], [41] [2_2:147] — OK: Lessard2016 p. 1 title: “A Bio-Inspired Tensegrity Manipulator with Multi-DOF, Structurally Compliant Joints” — direct match. Fasquelle2020 (V25) introduces the 3-DoF X-joint manipulator with axial-only-loaded bars. [Lessard2016 PDF](#) — [Fasquelle2020 PDF](#)
- **V28** — [43], [40] [2_2:151-152] — OK: Klein2023 p. 1 verbatim: **“Arbeitsraumanalyse, Simulation und Bewegungsplanung eines seilgetriebenen robotischen Manipulators ... Friedrich-Alexander-Universität Erlangen-Nürnberg, Lehrstuhl für Fertigungsautomatisierung”** — confirms FAPS-internal Masterarbeit attribution. Walter2023 conclusion p. 354/ chapter-p. 350 verbatim: **“isolates impact forces up to 83 %”** (cf. V22); the experimental setup (modal hammer applied vertically to joint body, base connected to ground) is described p. 350/ chapter-p. 346. [Klein2023 PDF](#) — [Walter2023 PDF](#)
- **V29** — [39], [44] [2_2:156-157] — OK: Bicchi2004 (V22). Pratt1995 (“Series Elastic Actuators”, MIT AI Lab) p. 1 abstract: **“It is traditional to make the interface between an actuator and its load as stiff as possible ... reducing interface stiffness offers a number of advantages, including greater shock tolerance, lower reflected inertia, more accurate and stable force control ... As a trade-off, reducing interface stiffness also lowers zero motion force bandwidth”** — direct support for SEA bandwidth/shock-tolerance trade-off. [Bicchi2004 PDF](#) — [Pratt1995 PDF](#)

8.3 §2.3 Workspace Analysis

- **V1** — [45], [46] [2_3:10] — OK: framing claim that workspace analysis verifies reachability and exposes singular/poor-quality regions. Rastegar1990 (MMT 25(2):233–239) p. 1 §1 “Introduction”: “The workspace of manipulators has been studied using analytical and numerical techniques ... extreme reach ... boundary surfaces have been investigated”; Dong2013 (MMT 70:508–522) p. 1 abstract: “reachable workspaces of robotic manipulators ... sampling joint angles and evaluating the boundary of the resulting set in the space of rigid-body motions”. [Rastegar1990 PDF](#) — [Dong2013 PDF](#)
- **V2** — [45] [2_3:18] — OK: Monte Carlo motivation. p. 1 abstract verbatim: “The Monte Carlo method is shown to be very simple, and readily implemented”; p. 1 §1: “probabilistic methods have not been applied to the problems of analysis and synthesis of manipulators” — supports the “no explicit boundary derivation” rationale. [PDF](#)
- **V3** — [47] [2_3:22] — OK (resolved 2026-05-02): MathWorks Robotics System Toolbox reference page `generateRobotWorkspace` (R2026a release; URL `/help/robotics/ref/generaterobotworkspace.html`) documents an API that returns reachable workspace point sets paired with joint configurations — directly matches the cited workflow. Local HTML snapshot now archived at `workspace_analysis/Web_Resources/mathworks_generaterobotworkspace.html`. [MathWorks generateRobotWorkspace snapshot](#)

- **V4** — **[46]** [2_3:24] — OK: voxel discretisation + density-based indicators. Dong2013 p. 4 verbatim: “We divide the workspace, W , into small boxes (voxels) of equal size $\Delta x = 0.15$, $\Delta y = 0.15$, $\Delta \theta = \frac{\pi}{4}$. The workspace density ρ assigns each box of the workspace W the number of points within the box that are reachable, normalized so as to be a probability density.” [PDF](#)
- **V5** — **[48]** [2_3:33, eq:yoshikawa_manipulability] — OK: equation $w = \sqrt{\det(JJ^\top)}$ matches the measure introduced in Yoshikawa1985 (IJRR 4(2):3–9). p. 2 verbatim: “ w is equal to the volume of the manipulability ellipsoid except for the constant coefficient d ”; p. 1 (singular state passage): “the manipulator is in a singular state. This state θ^* is not desirable because the manipulation vector r cannot move in a certain direction” — supports the “becomes zero at kinematic singularities” claim. [PDF](#)
- **V6** — **[37]** [2_3:36-41, eq:inv_condition_number] — OK: condition-number-as-isotropy index. Salisbury & Craig 1982 (IJRR 1(1):4–17) p. 6 ll. 18–33 verbatim: “the relative error is bounded by the product of the condition number of the Jacobian transpose matrix and the relative error in joint torque ... Points in the workspace that minimize the condition number of the Jacobian matrix are the best conditioned to minimize error propagation.” The inverse condition number $\kappa_{\text{inv}} = \frac{\sigma_{\min}}{\sigma_{\max}} \in [0, 1]$ is the standard reciprocal of $\text{cond}(J) = \frac{\sigma_{\max}}{\sigma_{\min}}$ (textbook identity, no symbol-level mismatch with the source). The “Salisbury and Craig introduced” attribution is consistent with Yoshikawa1985 p. 6 acknowledging “Craig (1982) have used the condition number ... as a measure of workspace quality”. [PDF](#)
- **V7** — **[46]** [2_3:44] — OK: reachability-density definition. Dong2013 p. 4 (cf. V4) defines density per voxel as the count of reachable points normalised to a probability density — direct match for the thesis’s “joint-space volume that maps into the voxel region” framing. [PDF](#)
- **V8** — **[48]**, **[37]**, **[46]** [2_3:48] — OK: implications-for-RL paragraph reuses V5/V6/V7 evidence (manipulability volume + condition number + reachability density) to motivate task-space placement; no new quantitative claim is added beyond what those three sources establish. [Yoshikawa1985 PDF](#) — [Salisbury1982 PDF](#) — [Dong2013 PDF](#)
- **V9** — **[20]**, **[39]** [2_3:50] — OK: tendon/tensegrity-compliance-amplifies-Cartesian-error claim. Miyasaka2020 (T-Mech 25(2):1095, 2020) hysteresis/cable-friction modelling — already verified in §2.2 V3. Bicchi2004 (passive-compliance / contact-force argument) — already verified in §2.2 V22; the implication that compliance can amplify positional uncertainty into contact forces follows from the same modelling stance. [Miyasaka2020 PDF](#) — [Bicchi2004 PDF](#)

8.4 §2.4 Physics Simulation

- **V1** — **[49]**, **[50]** [2_4:9] — OK: USD-physics-schema layering claim. [openusd_usdphysics_schema.html](#) (local snapshot under `simulation/Documentation/`) page title verbatim: “UsdPhysics: USD Physics Schema — OpenUSD Documentation”. [omni_physx_overview.html](#) body: “The omni.physx extension provides a connection between USD physics content and the NVIDIA PhysX simulation engine. Physics content in USD is defined through two USD schemas: UsdPhysics ... and PhysXS[chema]”. [openusd_usdphysics_schema](#) — [omni_physx_overview](#)
- **V2** — **[50]**, **[51]**, **[52]**, **[53]** [2_4:11] — OK: omni.physx parses USD + PhysX 5 backend + GPU pipeline claim. [omni_physx_overview.html](#) documents the USD↔PhysX connection (cf. V1). [physx_541_gpu_rigid_bodies.html](#) body verbatim: “PhysX simula-

tion can be configured to take advantage of CUDA capable GPUs under Linux or Windows. ... It supports the entire rigid body feature-set including articulation[s]”. [isaac_sim_510_physics_fundamentals.html](#) and [physx_welcome_513.html](#) confirm Isaac Sim 5.1 / PhysX 5 stack. [omni_physx_overview](#) — [isaac_sim_510_physics_fundamentals](#) — [physx_welcome_513](#) — [physx_541_gpu_rigid_bodies](#)

- **V3** — **[54], [55]** [2_4:13] — OK: reality-gap framing. Jakobi1995 (SAB 1995, Sussex) p. 1 title verbatim: “Noise and The Reality Gap: The Use of Simulation in Evolutionary Robotics” (PDF text-layer is letter-spaced; title established from filename + p. 1 header); p. 3 body: “system robust enough to cope with such an envelope-of-noise may handle the transfer from simulation to reality better”. Salvato2021 (IEEE Access, DOI 10.1109/ACCESS.2021.3126658) p. 1 title verbatim: “Crossing the Reality Gap: A Survey on Sim-to-Real Transferability of Robot Controllers in Reinforcement Learning”. [Jakobi1995 PDF](#) — [Salvato2021 PDF](#)
- **V4** — **[55], [13]** [2_4:13] — OK: simulator-as-data-generator + transition-distribution claim. Salvato2021 abstract (cf. V3) frames the survey around RL controllers crossing the reality gap. Peng2018 p. 4 §IV.C verbatim: “Dynamics Randomization. During training, rollouts are organized into episodes ... a random set of dynamics parameters μ are sampled according to ρ_μ and held fixed for the duration of the episode” — directly supports the modelling-shapes-the-transition-distribution claim. [Salvato2021 PDF](#) — [Peng2018 PDF](#)
- **V5** — **[13], [56]** [2_4:13] — OK: solver/timestep/contact choices as part of the learning-system design. Peng2018 p. 7 conclusion: “Training policies with randomized dynamics in simulation enables the resulting policies to be deployed directly on a physical robot despite poor calibrations”. [physx_best_practices_541.html](#) documents stiffness/timestep/contact-tuning recommendations directly. [Peng2018 PDF](#) — [physx_best_practices_541](#)
- **V6** — **[52], [57]** [2_4:23] — OK: rigid-body actors/shapes/materials description. [physx_welcome_513.html](#) is the PhysX 5.1.3 docs entry-point covering the whole rigid-body feature-set. [physx_rigid_body_dynamics_511.html](#) body lists “Actors Kinematic Surface Velocities Active Actors Dominance Solver Iterations ... Joints Articulations GPU Rigid Bodies” — directly matches the static/kinematic/dynamic-actor framing. [physx_welcome_513](#) — [physx_rigid_body_dynamics_511](#)
- **V7** — **[58]** [2_4:26] — OK: reduced-coordinates articulation claim. [physx_articulations_513.html](#) TOC verbatim: “Reduced-Coordinates and Comparison to Rigid Bodies” — direct match for the cited reduced-coordinates formulation and the mass-ratio / long-chain stability rationale. [physx_articulations_513](#)
- **V8** — **[58], [59]** [2_4:29] — OK: fixed/spatial articulation tendons. [physx_articulations_513.html](#) TOC: “Articulation Tendons / Fixed Tendons / Spatial Tendons”. [omni_physics_tendon_authoring.html](#) verbatim: “Creating a Fixed Tendon ... the tendon cannot link arbitrary joints in the articulation, but must follow the articul[at]ion structure]” and “Spatial tendons are created by adding attachments to articulation links”. [physx_articulations_513](#) — [omni_physics_tendon_authoring](#)
- **V9** — **[60]** [2_4:35] — OK: contact-as-constrained-dynamics + interpenetration prevention. [physx_rigid_body_collision_510.html](#) is the PhysX 5.1.0 rigid-body-collision docs page covering contact generation, restitution, friction. [physx_rigid_body_collision_510](#)
- **V10** — **[57], [61]** [2_4:35] — OK: friction/restitution via materials with static + dynamic coefficients. [physx_material_api_510.html](#) is the PhysX material API page (static/dynamic friction + restitution). [physx_rigid_body_dynamics_511](#) — [physx_material_api_510](#)

- **V11** — [55], [13] [2_4:35] — OK: contact-rich-grasping + sim-to-real-robustness coupling — Salvato2021 survey (cf. V3) and Peng2018 dynamics-randomisation (cf. V4) jointly support this; no new quantitative claim. [Salvato2021 PDF](#) — [Peng2018 PDF](#)
- **V12** — [57] [2_4:38] — OK: PGS vs TGS solver claim. [physx_rigid_body_dynamics_511.html](#) body verbatim: “PhysX 5.1 the new default solver is the Temporal Gauss-Seidel (TGS) solver. As an alternative and for compatibility the previously-default Projected Gauss-Seidel-style (PGS) solver is still available. Temporal gauss-seidel offers several advantages over the PGS-style solver: Dramatically improved convergence”. [physx_rigid_body_dynamics_511](#)
- **V13** — [62], [56] [2_4:38] — OK: smaller substeps > more iterations for stiff systems. Macklin2019 (NVIDIA) p. 1 verbatim: “performing a single large time step with n constraint solver iterations is less effective than computing n smaller time steps, each with a single constraint solver iteration”; “split every visual time step into n substeps of length $\Delta \frac{t}{n}$ and ... perform a single iteration of extended position-based dynamics (XPBD)”. [Macklin2019 PDF](#) — [physx_best_practices_541](#)
- **V14** — [63] [2_4:51] — OK (table row “Time step h ”): [physx_simulation_541.html](#) documents the simulation-stepping API and time-step trade-offs. [physx_simulation_541](#)
- **V15** — [57] [2_4:52] — OK (table row “Solver iterations”): cf. V12; PhysX 5.1.1 rigid-body docs document position/velocity solver-iteration counts. [physx_rigid_body_dynamics_511](#)
- **V16** — [64] [2_4:53] — OK (table row “contactOffset/restOffset”): [physx_advanced_collision_detection_512.html](#) is the canonical PhysX 5.1.2 advanced-collision-detection docs page covering the contact/rest-offset parameters. [physx_advanced_collision_detection_512](#)
- **V17** — [61] [2_4:54] — OK (table row “Friction coefficients”): cf. V10. [physx_material_api_510](#)
- **V18** — [63], [56] [2_4:61] — OK: time-step accuracy/stability discussion. [physx_simulation_541.html](#) (timestep + substepping) and [physx_best_practices_541.html](#) (best-practice recommendations on timestep selection) jointly support the claim. [physx_simulation_541](#) — [physx_best_practices_541](#)
- **V19** — [60], [65] [2_4:63] — OK: collider-cost ranking (primitives < convex meshes). [physx_geometry_541.html](#) is the PhysX 5.4.1 geometry/shape docs page that documents primitive vs convex/triangle-mesh trade-offs; [physx_rigid_body_collision_510.html](#) covers shape-attached collision. [physx_rigid_body_collision_510](#) — [physx_geometry_541](#)
- **V20** — [53], [66] [2_4:67] — OK: GPU pipeline + order-of-magnitude throughput claim. [physx_541_gpu_rigid_bodies.html](#) body verbatim: “PhysX simulation can be configured to take advantage of CUDA capable GPUs ... a performance benefit proportional to the arithmetic complexity of a scene”. Makoviychuk2021 (Isaac Gym, NeurIPS Datasets/Benchmarks 2021) p. 1 abstract verbatim: “communicate by directly passing data from physics buffers to PyTorch tensors without ever going through CPU bottlenecks. This leads to blazing fast training times for complex robotics tasks on a single GPU with 2-3 orders of magnitude improvements compared to conventional RL training”. [physx_541_gpu_rigid_bodies](#) — [Makoviychuk2021 PDF](#)

8.5 §2.5 NVIDIA Isaac Sim and IsaacLab

- **V1** — [67], [68] [2_5:9] — OK: integrated-Omniverse-stack + GPU-parallel-MDP-training framing. [isaac_sim_510_overview.html](#) body verbatim: “Isaac Sim is the simulation itself: a

high-fidelity GPU-based PhysX engine, capable of supporting multi-sensor RTX rendering at an industrial scale”. Mittal2025 p. 1 abstract verbatim: “We present Isaac Lab, the natural successor to Isaac Gym, which extends the paradigm of GPU-native robotics simulation into the era of large-scale multi-modal learning”. [isaac_sim_510_overview](#) — [Mittal2025 PDF](#)

- **V2** — **[67]** [2_5:13] — OK: USD-as-scene-representation + Omniverse-Kit-extension architecture. [isaac_sim_510_overview.html](#) is the canonical “What Is Isaac Sim?” v5.1.0 page that documents Isaac Sim as an Omniverse reference application with USD-based assets and Kit extensions (cf. V1 quote on the GPU-PhysX/RTX stack). [isaac_sim_510_overview](#)
- **V3** — **[67], [69]** [2_5:22] — OK (resolved 2026-05-02): PhysX 5 GPU-accelerated rigid-body and articulation backend. PhysX 5.1.0 page ([physx_510_gpu_rigid_bodies.html](#), version-matched to bib URL, snapshotted 2026-05-02): “GPU Rigid Bodies — physx 5.1.0 documentation” — covers GPU-accelerated rigid-body feature-set including articulations. [isaac_sim_510_overview.html](#) confirms “high-fidelity GPU-based PhysX engine” (cf. V1). The 5.4.1 snapshot is retained for §2.4 cross-references. [isaac_sim_510_overview](#) — [physx_5.1.0_gpu_rigid_bodies](#)
- **V4** — **[70]** [2_5:23] — OK (resolved 2026-05-02): the Isaac Sim 5.1.0 Python Extensions API root page is now snapshotted locally at [isaac_sim_510_extensions_api.html](#) (page title verbatim “Isaac Sim: Extensions API”; the snapshot contains 410+ references to extensions including `isaacsim.core.cloner` for environment cloning). The cloner-tutorial snapshot ([isaacsim_cloner_tutorial.html](#)) is retained as supplementary evidence. [isaac_sim_510_extensions_api](#) snapshot — [isaacsim_cloner_tutorial](#)
- **V5** — **[67]** [2_5:24] — OK: USD-centric asset pipeline + scalable instancing. Same “What Is Isaac Sim?” page as V1/V2; documents USD assets (Nova Carter, SimReady USD assets) as the scene representation. [isaac_sim_510_overview](#)
- **V6** — **[68], [66], [71]** [2_5:28] — OK: Isaac Gym → Orbit → IsaacLab lineage. Mittal2025 p. 1 abstract verbatim “natural successor to Isaac Gym” (cf. V1). Makoviychuk2021 p. 1 title verbatim: “Isaac Gym: High Performance GPU Based Physics Simulation For Robot Learning”. Mittal2023 p. 1 abstract verbatim: “We present ORBIT, a unified and modular framework for robot learning powered by NVIDIA Isaac Sim”. [Mittal2025 PDF](#) — [Makoviychuk2021 PDF](#) — [Mittal2023 PDF](#)
- **V7** — **[72]** [2_5:31] — OK: manager-based architecture with reusable action/observation/event managers extended by reward/termination/command/curriculum components. [isaacsim_manager_based_rl_env.html](#) verbatim: “build a reinforcement learning task environment by extending Isaac Lab’s base environment with task-specific components”; “leverages the ManagerBasedRLEnv class, which extends the base environment with RL-specific functionality. Rather than modifying the core class directly, users implement a configuration object (ManagerBasedRLEnvCfg)”. [isaacsim_manager_based_rl_env](#)
- **V8** — **[72]** [2_5:33] — OK: ManagerBasedRLEnv returns MDP-relevant signals and is compatible with `gymnasium.make`. Same snapshot as V7 documents the ManagerBasedRLEnv class (verbatim quote in V7); compatibility with the Gymnasium interface is the standard Isaac Lab env-creation pattern documented across the manager-based env tutorial chain. [isaacsim_manager_based_rl_env](#)
- **V9** — **[66], [68], [73]** [2_5:36] — OK: GPU-resident pipeline + order-of-magnitude throughput + IsaacLab multi-GPU/multi-node extension. Makoviychuk2021 p. 1 abstract verbatim: “communicate by directly passing data from physics buffers to PyTorch tensors without ever going through CPU bottlenecks. This leads to blazing fast training times for complex robotics

tasks on a single GPU with 2-3 orders of magnitude improvements compared to conventional RL training”. Mittal2025 p. 2 verbatim: “these capabilities scale efficiently across multi-GPU and multi-node setups”. [isaacsim_multi_gpu_doc.html](#) body verbatim: “Isaac Lab supports multi-GPU and multi-node reinforcement learning. Currently, this feature is only available for RL-Games, RSL-RL and skrl libraries workflows”. [Makoviychuk2021 PDF](#) — [Mittal2025 PDF](#) — [isaacsim_multi_gpu_doc](#)

- **V10** — [\[67\]](#), [\[68\]](#) [2_5:47] — OK (table row “Isaac Sim + IsaacLab — Integrated Omniverse stack: PhysX, RTX, RL tooling”): supported by the same evidence as V1 (“high-fidelity GPU-based PhysX engine ... multi-sensor RTX rendering” + Mittal2025 abstract on the manager-based composable architecture). [isaacsim_510_overview](#) — [Mittal2025 PDF](#)
- **V11** — [\[74\]](#) [2_5:48] — OK (table row “MuJoCo — Efficient dynamics for model-based control”): Todorov2012 p. 1 verbatim title “MuJoCo: A physics engine for model-based control” and abstract “We describe a new physics engine tailored to model-based control. Multi-joint dynamics are represented in generalized coordinates and computed via recursive algorithms”. [Todorov2012 PDF](#)
- **V12** — [\[75\]](#) [2_5:49] — OK (table row “PyBullet/Bullet — Accessible real-time physics”): local snapshot [pybullet_org.html](#) is the canonical Bullet/PyBullet project homepage entry as cited in [literature.bib](#) (URL <https://pybullet.org/wordpress/>). [pybullet_org snapshot](#)
- **V13** — [\[74\]](#), [\[75\]](#), [\[67\]](#), [\[68\]](#) [2_5:55] — OK: closing-paragraph qualitative comparison reuses V11/V12/V1/V10 evidence (MuJoCo as efficient per-step continuous-control engine; PyBullet as Python-accessible prototyping; Isaac Sim+IsaacLab as integrated USD/PhysX/MDP-training stack). No new quantitative claim. [Todorov2012 PDF](#) — [pybullet_org snapshot](#) — [isaacsim_510_overview](#) — [Mittal2025 PDF](#)

8.6 §2.6 Robot and Scene Description Formats

- **V1** — [\[76\]](#) [2_6:13] — OK (resolved 2026-05-02): prose rewritten to put composition first and interchange second, matching the FAQ §“Isn’t USD just another file format?” framing verbatim: “While USD does provide interchange services (more below), its primary function is to compose together scene description spread across many files (or other data sources) in many different formats ... to present a single hierarchical scenegraph view of the data.” Thesis now reads: “*USD is a system for composing scalable, hierarchically organised scene description from many layers and assets, and additionally provides interchange services across digital content creation applications.*” [openusd_usd_faq](#)
- **V2** — [\[76\]](#) [2_6:13] — OK: composition / sparse-overrides / references / variants. FAQ §“What is USD?” verbatim: “aggregate multiple assets into a single scenegraph while still allowing for sparse overrides”; §“non-destructive”: “shared between instances and scenes, and retain its integrity thanks to USD’s many non-destructive editing features”. [openusd_usd_faq](#)
- **V3** — [\[49\]](#) [2_6:15] — OK: UsdPhysics schemas for rigid bodies, joints, articulations. Local snapshot verbatim: “Joint types: Fixed, Revolute, Prismatic, Spherical, Distance, and D6 (generic). Articulations: Support reduced-coordinate systems”; “extends Universal Scene Description to represent real-time physics data, enabling rigid body simulations”. [openusd_usdphysics_schema](#)
- **V4** — [\[77\]](#) [2_6:15] — OK: simulator-specific PhysXSchema additions (PhysX joint parameters layered on UsdPhysics joints). Snapshot verbatim: “It can be applied to a UsdPhysics.Joint prim or the derived classes UsdPhysics.RevoluteJoint,

UsdPhysics.PrismaticJoint, or UsdPhysics.SphericalJoint. For overlapping parameters with PhysxJointAPI, the parameters of PhysxJointAxisAPI ...”; PhysxJointAxisAPI applies to “joints that are part of an articulation”. [omni_physics_physx_joint_schema](#)

- **V5** — **[78]** [2_6:19] — OK: URDF as XML-based ROS format for robot geometry/organization. Snapshot title verbatim: “URDF — ROS 2 Documentation: Rolling documentation”; the page is the canonical ROS 2 entry-point for the URDF tutorial chain (Building a visual / movable robot model, physical/collision properties, Xacro). [ros2_urdf_main](#)
- **V6** — **[79]** [2_6:19] — OK: collision geometry, inertial parameters, joint limits/dynamics on links. Snapshot verbatim: “to specify the center of gravity and the inertial reference frame (relative to the link’s reference frame). When using realtime controllers, inertia elements of zero (or almost zero) can cause the robot model to collapse”; TOC: “Collision / Physical Properties / Inertia / Contact Coefficients / Joint Dynamics”. [ros2_urdf_physical_properties](#)
- **V7** — **[80]** [2_6:22] — OK: closed-loop coupling handled outside standard URDF; mimic-joint linear coupling. The `hardware_interface` Joint-Kinematics page documents `MimicJoint` as a first-class struct (“Struct `MimicJoint`”); `ros2_control` treats mimic relationships as transmission/joint-coupling metadata rather than URDF kinematic links. *Note*: the page itself is an API index; for richer prose-evidence on mimic semantics, see the `omni_physics_current_limitations.html` “Fixed tendons: Use the Mimic Joint feature” line (cf. V11). [ros2_control_joint_kinematics](#)
- **V8** — **[76], [49]** [2_6:33] — OK (table row “USD — World assembly, physics/rendering schemas”): reuses V1/V2 (USD composition + sparse overrides) and V3 (UsdPhysics schemas) evidence. [openusd_usd_faq](#) — [openusd_usdphysics_schema](#)
- **V9** — **[78], [81]** [2_6:34] — OK (table row “URDF — Robot model interchange; converted into USD”): [ros2_urdf_main](#) as canonical URDF reference (cf. V5); `isaacsim_import_urdf_510.html` is the Isaac Sim “Tutorial: Import URDF” page documenting the URDF-to-USD importer (page title verbatim: “Tutorial: Import URDF — Isaac Sim Documentation”; body shows `import_config.fix_base = True, make_default_prim = True` — the URDF import-config API). [ros2_urdf_main](#) — [isaacsim_import_urdf_510](#)
- **V10** — **[81], [82], [49], [77]** [2_6:40] — OK: URDF-to-USD import + IsaacLab “convert to USD first” recommendation + UsdPhysics/PhysXSchema attachment. `isaacsim_import_new_asset.html` verbatim: “NVIDIA Omniverse relies on the Universal Scene Description (USD) file format” — the page documents `Using URDF Importer / MJCF Importer / Mesh Importer` as the canonical asset-onboarding path. [isaacsim_import_urdf_510](#) — [isaacsim_import_new_asset](#) — [openusd_usdphysics_schema](#) — [omni_physics_physx_joint_schema](#)
- **V11** — **[83]** [2_6:43] — OK: PhysX articulation tendons listed under known limitations. Snapshot Table 1 “PhysX Engine Limitations” verbatim: “Articulation Tendons / The simulation fidelity and behavior can be inadequate. Articulation joint incoming force reported will be excessively high when using nonzero TGS velocity iterations. Fixed tendons: Use the Mimic Joint feature. Spatial tendons: Apply external forces to the links to mimic the spatial tendons.” [omni_physics_current_limitations](#)
- **V12** — **[83], [80]** [2_6:43] — OK: mimic-joint coupling for linearly coupled tendon effects + external-force computation for configuration-dependent tendon geometry. Same Table 1 row as V11 documents both workarounds verbatim (“Fixed tendons: Use the Mimic Joint feature” / “Spatial tendons: Apply external forces”); [ros2_control_joint_kinematics](#) provides

the mimic-joint API on the ROS side (Struct MimicJoint). [omni_physics_current_limitations](#) — [ros2_control_joint_kinematics](#)

8.7 §2.6 Prior Work (research-gap framing)

- **V1** — **[66]** [PW: 12] — OK: 2–3-orders-of-magnitude GPU speedup + Franka cube stacking + Shadow Hand cube rotation. Makoviychuk2021 p. 1 abstract verbatim: “blazing fast training times for complex robotics tasks on a single GPU with 2-3 orders of magnitude improvements compared to conventional RL training”. *Note:* the specific “16,384 parallel agents / under 25 min on A100 / 1 h Shadow Hand” timings paraphrase the paper’s benchmarks section but were not literally re-quoted this pass; canonical claim is supported by abstract + benchmark figures. [Makoviychuk2021 PDF](#)
- **V2** — **[84]** [PW: 15] — OK (resolved 2026-05-02): contact-reduction figures retained, speedup phrasing softened. Narang2022 p. 2 verbatim: “a novel synthesis of signed distance function (SDF)-based collisions, contact reduction, and a Gauss-Seidel solver”; “A naive contact generation scheme between a nut and bolt may generate $\sim 1e6$ contacts; a careful one may generate $\sim 1e4$ ” — supports the “ $\approx 1M \rightarrow \approx 10,000$ ” claim. The earlier “20,000-fold speedup” wording (which paraphrased a $7e4\times$ figure attributed to prior work [106] vs Bullet 2012, not to Factory itself) has been replaced with “order-of-magnitude speedups over prior contact simulation”. [Narang2022 PDF](#)
- **V3** — **[85]** [PW: 15] — OK: 83–99% / 600 trials / 0.5–0.6 mm clearance / SAPU + SBC. Tang2023 p. 2 verbatim: “of our simulation-trained policies, with success rates of 83-99% over 600 trials”; p. 2: “We also use realistic part clearances (≤ 0.5 - 0.6 mm, aligned with the upper bound of ISO 286)”; p. 1 abstract: “simulation-aware policy update (SAPU) ... and a sampling-based curriculum (SBC) to prevent overfitting to earlier stages of a curriculum”. [Tang2023 PDF](#)
- **V4** — **[86]** [PW: 18] — OK: 83% real-world success / 6-DoF cube reposing / asymmetric actor-critic / keypoint representation / 16,384 parallel envs on a single GPU. Allshire2022 p. 1 abstract verbatim: “we achieve a high success rate of 83% on a real TriFinger system, with a single policy able to perform grasping, ungrasping, and finger gaiting in order to achieve arbitrary poses within the workspace”; p. 3 Fig. 3: “Our system trains using the IsaacGym simulator on 16,384 environments in parallel on a single NVIDIA Tesla V100 or RTX 3090 GPU”; Table I: “Asymmetric actor-critic to learn dexterous manipulation”. [Allshire2022 PDF](#)
- **V5** — **[87]** [PW: 18] — OK: agile in-hand Allegro Hand + Automatic Domain Randomization. Handa2023 p. 1 title verbatim: “DeXtreme: Transfer of Agile In-hand Manipulation from Simulation to Reality”; p. 7 verbatim: “we set the parameters of the domain randomisations via a vectorised implementation of Automatic Domain Randomisation (ADR ...). ADR automatically adjus[ts ranges]”. [Handa2023 PDF](#)
- **V6** — **[86], [87]** [PW: 18] — OK: PPO + asymmetric-actor-critic shared by both; Allshire2022 Table I documents asymmetric critic observations (cf. V4); Handa2023 reuses the same training paradigm (the DeXtreme system explicitly adopts ADR + PPO atop Isaac Gym). [Allshire2022 PDF](#) — [Handa2023 PDF](#)
- **V7** — **[71]** [PW: 21] — OK: Orbit as modular Isaac-Sim-powered framework wrapping multiple RL libraries with benchmark tasks. Mittal2023 p. 1 abstract verbatim: “ORBIT, a unified and modular framework for robot learning powered by NVIDIA Isaac Sim. It offers a modular design to easily and efficiently create robotic environments with photo-realistic scenes and

high-fidelity rigid and deformab[le]. *Note*: “20+ benchmark tasks” + “wrapping four RL libraries (rl_games, RSL-RL, Stable-Baselines3, skrl)” not re-quoted verbatim this pass but is the paper’s documented integration list. [Mittal2023 PDF](#)

- **V8 — [68]** [PW:21] — OK (resolved 2026-05-02): Mittal2025 p. 20 verbatim: “the Franka cabinet task reaches over 1.6 million frames per second”; p. 2 verbatim: “filtered contact reporting, mimic joint systems, closed-loop kinematic chains, deformable objects (cloth and soft bodies), and coupled solvers for rigid and deformable bodies” — supports the mimic-joint / closed-loop / deformable-solver claims. The earlier “March 2024” release-date qualifier (which is not stated in the paper) has been removed from the prose. [Mittal2025 PDF](#)
- **V9 — [88]** [PW:21] — OK: skrl — modular RL library with native Isaac Lab support. SerranoMunoz2023 p. 1 title verbatim: “skrl: Modular and Flexible Library for Reinforcement Learning” (JMLR 24, 2023); p. 2 frames the library around “the reusability of its constituent components”. The “seven standalone components” / “first-class Isaac Lab support” wording is paraphrased from §1–§3; the modular-components claim is fully supported. [SerranoMunoz2023 PDF](#)
- **V10 — [89]** [PW:26] — OK (resolved 2026-05-02): 9-tendon, 18-DoF tendon-driven manipulator + curriculum-RL in MuJoCo. Or2023 p. 5 Table 2 (action space) verbatim: “*Dorsal tendon (1D) ... Ventral tendons (8D) ... Joint angles (18D)*”— confirms $1 + 8 = 9$ tendons and 18-DoF joint observation. p. 3 verbatim: “*The links of the Robostrich arm were numbered from the C2 to C18, corresponding in order from the cranial to the caudal part of the ostrich vertebra*”; p. 4 verbatim: “*The Robostrich arm comprises 18 rigid links with passive rotational joints*”. p. 1 abstract verbatim: “*manipulator inspired from an ostrich neck referred to as the Robostrich arm ... We proposed a curriculum-based reinforcement-learning approac[h]*”; p. 2 mentions MuJoCo. The thesis prose previously claimed “24-cable, 15-DoF”; this has been corrected to “9-tendon, 18-DoF” to match the paper’s own action/observation table. [Or2023 PDF](#)
- **V11 — [90]** [PW:26] — OK: PAMY2 pneumatic-tendon arm + safe high-speed motion via learned dynamics. Guist2024 p. 1 verbatim: “Pairing our robot with pneumatic muscles allows generating high forces and highly accelerated motions, while benefiting from impact resilience through passive compliance”. *Note*: the explicit “model-based RL embedded in a trajectory optimizer” wording paraphrases the system architecture but was not literally re-quoted this pass. [Guist2024 PDF](#)
- **V12 — [89], [90]** [PW:26] — OK: both works rely on MuJoCo-native tendon elements with routing/moment-arm support — supported by the same source quotes as V10/V11; the MuJoCo tendon API is documented in Todorov2012 §V (cf. §2.5 V11). [Or2023 PDF](#) — [Guist2024 PDF](#)
- **V13 — [91]** [PW:26] — OK on the comparison, **attribution-author-order note**. Local PDF verbatim: “Characterization of continuum robot arms under reinforcement learning ... Ryota Morimoto, Masahiro Ikeda, Ryuma Niiyama and Yasuo Kuniyoshi” (Front. Robot. AI 9:895388). p. 1 abstract verbatim: “we examined the advantages of a continuum robot arm as compared to a typical and rigid seven-degree-of-freedom (7-DoF) robot manipulator”. The thesis already attributes the work to “Morimoto *et al.*” (correct first author) — bib key remains Saito2022ContinuumRL for legacy reasons. [Saito/Morimoto2022 PDF](#)
- **V14 — [92]** [PW:29] — OK: CaRoSaC — Unity3D + Obi Rope (XPBD) + TD3 controller for industrial 4-cable parallel robot. Dhakate2025 p. 1 verbatim: “CaRoSaC: A Reinforcement Learning-Based Kinematic Control of Cable-Driven Parallel Robots by Addressing Cable Sag through Simulation”; p. 3: “we utilized the Obi Rope extension from Virtual Method Studio for

Unity3D ... Its underlying solver is based on Position Based Dynamics”; p. 4: “Twin Delayed Deep Deterministic Policy Gradient algorithm (TD3) ... We trained a TD3 agent to learn a task-space position tracking controller”. [Dhakate2025 PDF](#)

- **V15** — **[93]** [PW:29] — OK: 0.27% peak position divergence vs hardware prototype across speed scenarios. Garrido2024 p. 2 verbatim: “the performance of its industrial counterpart, with a peak divergence of 0.27% in the position of the robot under various speed scenarios. This occurs even when accounting for latency arising from data exchange”. [Garrido2024 PDF](#)
- **V16** — **[15]** [PW:32] — OK (citation switched 2026-04-24, Lilge2021BenchmarkTDCR was a duplicate of Rao2021): the chapter now cites @Rao2021TDCRModeling directly and attributes the work to “Rao *et al.*”. The duplicate @article{Lilge2021BenchmarkTDCR, ...} entry in literature.bib (line 1411) is now unused and can be removed at user discretion. PDF locally available as tendon_robots/Review_Articles/Rao2021.pdf. [Rao2021 PDF](#)
- **V17** — **[38]** [PW:41] — OK (file present under filename Shah2022.pdf, verified 2026-04-24): tensegrity robotics survey by Shah, Booth, Baines, Wang, Vespignani, Bekris, Kramer-Bottiglio (Soft Robotics 9(4):639–656, 2022). Local PDF p. 1 verbatim: “Tensegrity Robotics / Dylan S. Shah, Joran W. Booth, Robert L. Baines, Kun Wang, Massimo Vespignani, Kostas Bekris, and Rebecca Kramer-Bottiglio”; abstract: “Numerous recent advances in robotics have been inspired by the biological principle of tensile integrity—or ‘tensegrity’—to achieve remarkable feats of dexterity and resilience.” [Shah2022 PDF \(= ShahTensegrityRobotics2022\)](#)
- **V18** — **[42]** [PW:41] — OK: bio-inspired tensegrity arm with structurally compliant joints. Lessard2016 p. 1 title verbatim: “A Bio-Inspired Tensegrity Manipulator with Multi-DOF, Structurally Compliant Joints”. [Lessard2016 PDF](#)
- **V19** — **[27]** [PW:41] — OK: planar tensegrity 2-X manipulator with antiparallelogram joints + kinetostatic analysis. Furet & Wenger 2019 p. 1 title verbatim: “Kinetostatic analysis and actuation strategy of a planar tensegrity 2-X manipulator”; p. 3: “antiparallelogram. The mechanism assembly condition is $b \leq L$ ”. [Furet2019 PDF](#)
- **V20** — **[41], [94]** [PW:41] — OK (Fasquelle2022 PDF acquired 2026-04-24): Fasquelle2020 p. 1 verbatim: “A bio-inspired 3-DOF light-weight manipulator with tensegrity X-joints”; “experiment results conducted on a test-bed prototype are presented”. Fasquelle2022 (HAL/JMR preprint, filed locally as Fasquelle2022.pdf) p. 1 verbatim: “Identification and control of a 3-X cable-driven manipulator inspired from the bird neck” — supports the system-identification + closed-loop position control claim. [Fasquelle2020 PDF](#) — [Fasquelle2022 PDF](#)
- **V21** — **[18]** [PW:41] — OK: antagonistically-actuated R-joint vs X-joint comparison, workspace + stiffness modulation + singularities. Muralidharan & Wenger 2021 p. 1 verbatim: “Optimal design and comparative study of two antagonistically actuated tensegrity joints”; “the R-joint has a fixed center of rotation while the X-joint has a moving center of rotation. Both joints are equipped with two lateral springs and are actuated by cables”. [Muralidharan2021 PDF](#)
- **V22** — **[95]** [PW:44] — OK (PDF acquired 2026-04-24, Yoon2021DLRWrist citation dropped): Hamon & Aoustin 2010 (Humanoids 2010, doi 10.1109/ichr.2010.5686851) PDF locally available. p. 1 verbatim: “Cross four-bar linkage for the knees of a planar bipedal robot ... A. Hamon and Y. Aoustin, IRCCyN, UMR 6597, Ecole Centrale de Nantes”. The Yoon2021DLRWrist mention was dropped from the chapter (PDF was paywalled and the example was non-essential). *Note:* the bib booktitle field for HamonAoustin2010 claims IROS

2010 but the paper is from Humanoids 2010 — user may want to fix this in `literature.bib`. [HamonAoustin2010 PDF](#)

- **V23 — [18]** [PW:44] — OK (Muralidharan2024b citation dropped 2026-04-24): the workspace-comparison sentence now folds into the existing MuralidharanWenger2021 citation, which already covers 2-X vs 2-R workspace coverage / stiffness modulation / singularity comparison (see also §2.2 V21 / PW V21). The bare `@Muralidharan2024b` (no bib entry, no PDF) was removed from the chapter. [Muralidharan2021 PDF](#)
- **V24 — [96]** [PW:48] — OK: model-based RL (MDGPS) on SUPERball tensegrity **locomotion**. Zhang2017 p. 1 title verbatim: “Deep Reinforcement Learning for Tensegrity Robot Locomotion”; abstract: “a novel extension of mirror descent guided policy search (MDGPS) applied to periodic locomotion movements ... evaluate our method with real-world and simulated experiments on the SUPERball tensegrity robot”. [Zhang2017tensRL PDF](#)
- **V25 — [40]** [PW:51] — OK: 83% impact-force isolation by the cable-driven tensegrity-inspired joint. Local source is the CableCon 2023 proceedings volume (Walter2023.pdf, 457 pp.); table-of-contents p. 11 lists the chapter “Tensegrity-Inspired Joint Can Protect from Impacts by Isolating ... 344 / Jonas Walter, Lukas Rothfischer, Richard Stierstorfer, Takeru Nemoto, Jörg Franke, and Sebastian Reitelshöfer”; p. 348 verbatim: “It was found that the joint isolates impact forces up to 83%, demonstrating its ability to effectively absorb and dissipate energy from external impacts”. [Walter2023 PDF \(proceedings p.~344\)](#)
- **V26 — [43]** [PW:51] — OK: FAPS Masterarbeit “Arbeitsraumanalyse, Simulation und Bewegungsplanung eines seilgetriebenen robotischen Manipulators” (FAU Erlangen-Nürnberg). Klein2023 p. 1 (cover) verbatim: “Arbeitsraumanalyse, Simulation und Bewegungsplanung eines seilgetriebenen robotischen Manipulators / Masterarbeit ... Friedrich-Alexander-Universität Erlangen-Nürnberg / Lehrstuhl für Fertigungsautomatisierung und Produktionssystematik”. [Klein2023 PDF](#)
- **V27 — [48]** [PW:56] — OK: $w = \sqrt{\det(JJ^T)}$ + manipulability ellipsoid + singular configurations. Already verified as §2.3 V5; same equation transcription holds. [Yoshikawa1985 PDF](#)
- **V28 — [97]** [PW:56] — OK (resolved 2026-05-02): Vahrenkamp2015 p. 1 abstract verbatim: “incorporates constraining factors, such as joint limits or the self-distance between manipulator and other parts of the robot. Based on this measure we show how an extended capability representation of the robot’s workspace can be built”; p. 1 eq. (1) verbatim: $w = \sqrt{\det(JJ^T)} = s_1 s_2 \dots s_n$. The thesis prose has been amended to make explicit that the cable-tension-limit-as-joint-limit analogy is the present author’s extension of Vahrenkamp’s joint-limit penalty (now framed as “By analogy with their joint-limit penalty, this constraint-penalised approach can be adapted to tendon-driven manipulators by treating cable tension bounds in place of joint-position bounds; this analogy is the present author’s extension and is not stated in [97]”). [Vahrenkamp2015 PDF](#)
- **V29 — [45]** [PW:59] — OK: Monte-Carlo forward-kinematics workspace sampling + non-uniform-density limitation. Rastegar1990 p. 1 title verbatim: “MANIPULATOR WORKSPACE ANALYSIS USING THE MONTE CARLO METHOD”; p. 1: “numerical method for solving mathematical problems by means of random sampling”. The “denser near center, sparser at boundaries” wording is a known property of the method; supported by the paper’s own density discussion. [Rastegar1990 PDF](#)
- **V30 — [98]** [PW:59] — OK: Gaussian-Growth Monte-Carlo improvement that densifies boundary regions for high-DoF robots. Peidro2017 p. 1 title verbatim: “An improved Monte Carlo method based on Gaussian growth to calculate the workspace of robots”; abstract:

“uses the Gaussian distribution to densify and grow this seed workspace until the boundaries of the workspace are attained ... compared with previous Monte Carlo methods using a 10-degrees-of-freedom robot as a case study”. [Peidro2017 PDF](#)

- **V31** — (Cao2017WorkspaceTDCR citation dropped 2026-04-24) [PW:59] — the tendon-driven continuum-robot workspace sentence was removed from the chapter. The remaining workspace-determination paragraph (Rastegar1990 + Peidro2017) covers the methodological foundations needed at this point in the prior-work narrative; the TDCR-specific superposition-and-interference claim was not load-bearing for the gap argument and the source PDF (ASME, paywalled) was not available. The bib entry is retained for future use.
- **V32** — [\[66\]](#), [\[84\]](#), [\[85\]](#), [\[86\]](#), [\[87\]](#) [PW:64] — OK: closing-paragraph “first body of work” (GPU-accelerated RL for manipulation in Isaac Gym/Sim with sim-to-real demonstrated for contact-rich rigid-body robots) — reuses V1/V2/V3/V4/V5 evidence; no new claim. [Makoviychuk2021 PDF](#) — [Narang2022 PDF](#) — [Tang2023 PDF](#) — [Allshire2022 PDF](#) — [Handa2023 PDF](#)
- **V33** — [\[89\]](#), [\[90\]](#), [\[91\]](#), [\[92\]](#) [PW:64] — OK: closing-paragraph “second body” (tendon/cable-driven RL predominantly in MuJoCo or specialised engines, not Isaac Sim) — reuses V10/V11/V13/V14 evidence. [Or2023 PDF](#) — [Guist2024 PDF](#) — [Saito2022 PDF](#) — [Dhakate2025 PDF](#)
- **V34** — [\[27\]](#), [\[41\]](#), [\[18\]](#) [PW:64] — OK: closing-paragraph “third body” (tensegrity-inspired manipulators with antiparallelogram joints studied from a mechanism-design perspective) — reuses V19/V20/V21 evidence. [Furet2019 PDF](#) — [Fasquelle2020 PDF](#) — [Muralidharan2021 PDF](#)
- **V35** — [\[43\]](#), [\[88\]](#), [\[14\]](#) [PW:66] — OK: gap-closure paragraph (FAPS-internal Klein workspace-analysis predecessor + skrl-PPO training stack). Schulman2017 p. 1 title verbatim: “Proximal Policy Optimization Algorithms”; abstract: “novel objective with clipped probability ratios, which forms a pessimistic estimate (i.e., lower bound) of the performance of the policy”. [Klein2023 PDF](#) — [SerranoMunoz2023 PDF](#) — [Schulman2017PPO PDF](#)

8.8 Provenance & next-pass policy

These bullets were extracted from inline `// VERIFIED: comments in chapters/background/2_*.typ` (commit-state of 2026-04-22). After extraction, the inline comments were removed from the chapter source files for readability. The reusable per-section verification prompt that regenerates and refines this log line-by-line lives at `chapters/background/VERIFICATION_PROMPT_TEMPLATE.md`.

Bibliography

- [1] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, Second. The MIT Press, 2018. Accessed: Mar. 03, 2026. [Online]. Available: <https://mitpress.mit.edu/9780262039246/reinforcement-learning/>
- [2] J. Kober, J. A. Bagnell, and J. Peters, “Reinforcement Learning in Robotics: A Survey,” *The International Journal of Robotics Research*, vol. 32, no. 11, pp. 1238–1274, 2013, doi: [10.1177/0278364913495721](https://doi.org/10.1177/0278364913495721).
- [3] M. T. J. Spaan, “Partially Observable Markov Decision Processes,” *Reinforcement Learning: State-of-the-Art*. Springer, 2012. Accessed: Mar. 03, 2026. [Online]. Available: <https://www.st.ewi.tudelft.nl/mtjspaen/pub/Spaan12pomdp.pdf>
- [4] A. Y. Ng, D. Harada, and S. J. Russell, “Policy Invariance Under Reward Transformations: Theory and Application to Reward Shaping,” in *Proceedings of the Sixteenth International Conference on Machine Learning (ICML 1999)*, San Francisco, CA, USA: Morgan Kaufmann, 1999, pp. 278–287. Accessed: Mar. 03, 2026. [Online]. Available: <https://www.cs.utexas.edu/~shivaram/readings/b2hd-NgHR1999.html>
- [5] S. Devlin and D. Kudenko, “Dynamic Potential-Based Reward Shaping,” in *Proceedings of the 11th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2012, pp. 433–440.
- [6] I. Popov *et al.*, “Data-efficient Deep Reinforcement Learning for Dexterous Manipulation,” *arXiv preprint arXiv:1704.03073*, 2017.
- [7] R. Toro Icarte, T. Q. Klassen, R. Valenzano, and S. A. McIlraith, “Reward Machines: Exploiting Reward Function Structure in Reinforcement Learning,” *Journal of Artificial Intelligence Research*, vol. 73, pp. 173–208, 2022.
- [8] Y. Zhu *et al.*, “robosuite: A Modular Simulation Framework and Benchmark for Robot Learning,” *arXiv preprint arXiv:2009.12293*, 2020.
- [9] ARISE Initiative, “robosuite source code: `\texttt{pick_place.py}` – `\texttt{staged_rewards() method}`.” Accessed: May 02, 2026. [Online]. Available: https://github.com/ARISE-Initiative/robosuite/blob/master/robosuite/environments/manipulation/pick_place.py
- [10] J. Schulman, P. Moritz, S. Levine, M. I. Jordan, and P. Abbeel, “High-Dimensional Continuous Control Using Generalized Advantage Estimation,” *arXiv preprint arXiv:1506.02438*, 2015, [Online]. Available: <https://arxiv.org/abs/1506.02438>
- [11] D. Kalashnikov *et al.*, “QT-Opt: Scalable Deep Reinforcement Learning for Vision-Based Robotic Manipulation,” in *Conference on Robot Learning (CoRL)*, 2018.
- [12] J. Tobin, R. Fong, A. Ray, J. Schneider, W. Zaremba, and P. Abbeel, “Domain Randomization for Transferring Deep Neural Networks from Simulation to the Real World,” in *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2017. doi: [10.1109/IROS.2017.8202133](https://doi.org/10.1109/IROS.2017.8202133).
- [13] X. B. Peng, M. Andrychowicz, W. Zaremba, and P. Abbeel, “Sim-to-Real Transfer of Robotic Control with Dynamics Randomization,” in *2018 IEEE International Confer-*

- ence on Robotics and Automation (ICRA)*, 2018, pp. 3803–3810. doi: [10.1109/ICRA.2018.8460528](https://doi.org/10.1109/ICRA.2018.8460528).
- [14] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, “Proximal Policy Optimization Algorithms,” *arXiv preprint arXiv:1707.06347*, 2017, Accessed: Mar. 03, 2026. [Online]. Available: <https://arxiv.org/abs/1707.06347>
- [15] P. Rao, Q. Peyron, S. Lilge, and J. Burgner-Kahrs, “How to Model Tendon-Driven Continuum Robots and Benchmark Modelling Performance,” *Frontiers in Robotics and AI*, vol. 7, p. 630245, 2021, doi: [10.3389/frobt.2020.630245](https://doi.org/10.3389/frobt.2020.630245).
- [16] T. Nemoto, J. Walter, C. Bachmann, M. Gerlich, S. Reitelshöfer, and J. Franke, “Highly Dynamic 2-DOF Cable-Driven Robotic Wrist Based on a Novel Topology,” *IEEE Robotics and Automation Letters*, vol. 7, no. 2, pp. 5727–5734, 2022, doi: [10.1109/LRA.2022.3159816](https://doi.org/10.1109/LRA.2022.3159816).
- [17] H. Kobayashi, K. Hyodo, and D. Ogane, “On Tendon-Driven Robotic Mechanisms with Redundant Tendons,” *International Journal of Robotics Research*, vol. 17, no. 5, pp. 561–571, 1998.
- [18] V. Muralidharan and P. Wenger, “Optimal Design and Comparative Study of Two Antagonistically Actuated Tensegrity Joints,” *Mechanism and Machine Theory*, vol. 159, p. 104249, 2021, doi: [10.1016/j.mechmachtheory.2021.104249](https://doi.org/10.1016/j.mechmachtheory.2021.104249).
- [19] L.-W. Tsai and J.-J. Lee, “Kinematic Analysis of Tendon-Driven Robotic Mechanisms Using Graph Theory,” *ASME Journal of Mechanisms, Transmissions, and Automation in Design*, vol. 111, no. 1, pp. 59–65, 1989.
- [20] M. Miyasaka, M. Haghighipanah, Y. Li, J. Matheson, A. Lewis, and B. Hannaford, “Modeling Cable-Driven Robot With Hysteresis and Cable–Pulley Network Friction,” *IEEE/ASME Transactions on Mechatronics*, vol. 25, no. 2, pp. 1095–1104, 2020, doi: [10.1109/TMECH.2020.2973428](https://doi.org/10.1109/TMECH.2020.2973428).
- [21] T. Kato, I. Okumura, H. Kose, K. Takagi, and N. Hata, “Tendon-driven continuum robot for neuroendoscopy: validation of extended kinematic mapping for hysteresis operation,” *International Journal of Computer Assisted Radiology and Surgery*, vol. 11, no. 4, pp. 589–602, 2016, doi: [10.1007/s11548-015-1310-2](https://doi.org/10.1007/s11548-015-1310-2).
- [22] J. M. McCarthy and G. S. Soh, *Geometric Design of Linkages*, 2nd ed. Springer, 2011. doi: [10.1007/978-1-4419-7892-9](https://doi.org/10.1007/978-1-4419-7892-9).
- [23] J. J. Uicker, G. R. Pennock, and J. E. Shigley, *Theory of Machines and Mechanisms*, 5th ed. New York: Oxford University Press, 2017.
- [24] R. S. Hartenberg and J. Denavit, “A Kinematic Notation for Lower Pair Mechanisms Based on Matrices,” *Journal of Applied Mechanics (Transactions of the ASME)*, vol. 77, pp. 215–221, 1955.
- [25] E. A. Dijksman, “On the Cognates of the Antiparallelogram,” *ASME Journal of Engineering for Industry*, vol. 99, no. 3, 1977, doi: [10.1115/1.3439273](https://doi.org/10.1115/1.3439273).
- [26] H. Stachel, “Higher-Order Flexibility of a Bipartite Planar Framework,” *Acta Technica Napocensis – Series: Applied Mathematics and Mechanics*, vol. 43, pp. 159–167, 2000.

- [27] M. Furet and P. Wenger, "Kinetostatic Analysis and Actuation Strategy of a Planar Tensegrity 2-X Manipulator," *ASME Journal of Mechanisms and Robotics*, vol. 11, no. 6, p. 60904, 2019, doi: [10.1115/1.4044209](https://doi.org/10.1115/1.4044209).
- [28] A. Pott, *Cable-Driven Parallel Robots – Theory and Application*, vol. 120. in Springer Tracts in Advanced Robotics, vol. 120. Springer, 2018. doi: [10.1007/978-3-319-76138-1](https://doi.org/10.1007/978-3-319-76138-1).
- [29] R. Verhoeven, "Analysis of the Workspace of Tendon-based Stewart Platforms," Doctoral dissertation, 2004.
- [30] T. Bruckmann, A. Pott, and M. Hiller, "Calculating Force Distributions for Redundantly Actuated Tendon-Based Stewart Platforms," *Advances in Robot Kinematics*, 2008.
- [31] A. Pott, T. Bruckmann, and L. Mikelsons, "Closed-Form Force Distribution for Parallel Wire Robots," in *Computational Kinematics*, Springer, 2009, pp. 25–34. doi: [10.1007/978-3-642-01947-0_4](https://doi.org/10.1007/978-3-642-01947-0_4).
- [32] M. Gouttefarde and C. M. Gosselin, "Analysis of the Wrench-Closure Workspace of Planar Parallel Cable-Driven Mechanisms," *ASME Journal of Mechanical Design*, vol. 128, no. 1, pp. 264–272, 2006.
- [33] J. J. Craig, *Introduction to Robotics: Mechanics and Control*, 3rd ed. Pearson Prentice Hall, 2005.
- [34] R. M. Murray, Z. Li, and S. S. Sastry, *A Mathematical Introduction to Robotic Manipulation*. CRC Press, 1994.
- [35] P. Bosscher, A. T. Riechel, and I. Ebert-Uphoff, "Wrench-Feasible Workspace Generation for Cable-Driven Robots," *IEEE Transactions on Robotics*, vol. 22, no. 5, pp. 890–902, 2006, doi: [10.1109/TRO.2006.878967](https://doi.org/10.1109/TRO.2006.878967).
- [36] L.-W. Tsai, *Robot Analysis: The Mechanics of Serial and Parallel Manipulators*. Wiley, 1999.
- [37] J. K. Salisbury and J. J. Craig, "Articulated Hands: Force Control and Kinematic Issues," *The International Journal of Robotics Research*, vol. 1, no. 1, pp. 4–17, 1982, doi: [10.1177/027836498200100102](https://doi.org/10.1177/027836498200100102).
- [38] D. S. Shah *et al.*, "Tensegrity Robotics," *Soft Robotics*, vol. 9, no. 4, pp. 639–656, 2022, doi: [10.1089/soro.2020.0170](https://doi.org/10.1089/soro.2020.0170).
- [39] A. Bicchi and G. Tonietti, "Fast and ``Soft-Arm" Tactics: Dealing with the Safety-Performance Tradeoff in Robot Arms Design and Control," *IEEE Robotics & Automation Magazine*, vol. 11, no. 2, pp. 22–33, 2004, doi: [10.1109/MRA.2004.1310939](https://doi.org/10.1109/MRA.2004.1310939).
- [40] J. Walter, L. Rothfischer, R. Stierstorfer, T. Nemoto, J. Franke, and S. Reitelshöfer, "Tensegrity-Inspired Joint Can Protect from Impacts by Isolating," in *Cable-Driven Parallel Robots – Proceedings of the 6th International Conference on Cable-Driven Parallel Robots (CableCon 2023)*, in Mechanisms and Machine Science, vol. 132. Springer, 2023, pp. 344–355. doi: [10.1007/978-3-031-32322-5_28](https://doi.org/10.1007/978-3-031-32322-5_28).
- [41] B. Fasquelle, M. Furet, P. Khanna, D. Chablat, C. Chevallereau, and P. Wenger, "A Bio-Inspired 3-DOF Light-Weight Manipulator with Tensegrity X-Joints," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2020, pp. 5054–5060. doi: [10.1109/ICRA40945.2020.9196589](https://doi.org/10.1109/ICRA40945.2020.9196589).

- [42] S. Lessard *et al.*, “A Bio-Inspired Tensegrity Manipulator with Multi-DOF, Structurally Compliant Joints,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2016, pp. 5515–5520. doi: [10.1109/IROS.2016.7759811](https://doi.org/10.1109/IROS.2016.7759811).
- [43] M. Klein, “Arbeitsraumanalyse, Simulation und Bewegungsplanung eines seilgetriebenen robotischen Manipulators,” Masterarbeit, 2023.
- [44] G. A. Pratt and M. M. Williamson, “Series Elastic Actuators,” pp. 399–406, 1995, doi: [10.1109/IROS.1995.525827](https://doi.org/10.1109/IROS.1995.525827).
- [45] J. Rastegar and B. Fardanesh, “Manipulator workspace analysis using the Monte Carlo method,” *Mechanism and Machine Theory*, vol. 25, no. 2, pp. 233–239, 1990, doi: [10.1016/0094-114X\(90\)90124-3](https://doi.org/10.1016/0094-114X(90)90124-3).
- [46] H. Dong, Z. Du, and G. S. Chirikjian, “Workspace density and inverse kinematics for planar serial revolute manipulators,” *Mechanism and Machine Theory*, vol. 70, pp. 508–522, 2013, doi: [10.1016/j.mechmachtheory.2013.08.008](https://doi.org/10.1016/j.mechmachtheory.2013.08.008).
- [47] The MathWorks, Inc., “Generate reachable workspace of robot in environment (\texttt{generateRobotWorkspace}).” Accessed: May 02, 2026. [Online]. Available: <https://www.mathworks.com/help/robotics/ref/generaterobotworkspace.html>
- [48] T. Yoshikawa, “Manipulability of Robotic Mechanisms,” *The International Journal of Robotics Research*, vol. 4, no. 2, pp. 3–9, 1985, doi: [10.1177/027836498500400201](https://doi.org/10.1177/027836498500400201).
- [49] Pixar Animation Studios, “UsdPhysics: USD Physics Schema.” Accessed: Mar. 03, 2026. [Online]. Available: https://openusd.org/docs/api/usd_physics_page_front.html
- [50] NVIDIA, “Omni PhysX — Overview.” Accessed: Mar. 03, 2026. [Online]. Available: https://docs.omniverse.nvidia.com/kit/docs/omni_physics/latest/extensions/runtime/source/omni.physx/docs/index.html
- [51] NVIDIA, “Physics Simulation Fundamentals.” Accessed: Mar. 03, 2026. [Online]. Available: https://docs.isaacsim.omniverse.nvidia.com/5.1.0/physics/simulation_fundamentals.html
- [52] NVIDIA, “Welcome to PhysX — PhysX 5.1.3 Documentation.” Accessed: Mar. 03, 2026. [Online]. Available: <https://nvidia-omniverse.github.io/PhysX/physx/5.1.3/index.html>
- [53] NVIDIA, “GPU Simulation (PhysX 5.4.1 Documentation).” Accessed: Mar. 03, 2026. [Online]. Available: <https://nvidia-omniverse.github.io/PhysX/physx/5.4.1/docs/GPURigidBodies.html>
- [54] N. Jakobi, P. Husbands, and I. Harvey, “Noise and the Reality Gap: The Use of Simulation in Evolutionary Robotics,” in *Advances in Artificial Life (ECAL 1995)*, in Lecture Notes in Computer Science, vol. 929. Springer, 1995, pp. 704–720. doi: [10.1007/3-540-59496-5_337](https://doi.org/10.1007/3-540-59496-5_337).
- [55] E. Salvato, G. Fenu, E. Medvet, and F. A. Pellegrino, “Crossing the Reality Gap: A Survey on Sim-to-Real Transferability of Robot Controllers in Reinforcement Learning,” *IEEE Access*, vol. 9, pp. 153171–153187, 2021, doi: [10.1109/ACCESS.2021.3126658](https://doi.org/10.1109/ACCESS.2021.3126658).
- [56] NVIDIA, “Best Practices Guide — PhysX 5.4.1 Documentation.” Accessed: Mar. 03, 2026. [Online]. Available: <https://nvidia-omniverse.github.io/PhysX/physx/5.4.1/docs/BestPractices.html>

- [57] NVIDIA, “Rigid Body Dynamics — PhysX 5.1.1 Documentation.” Accessed: Mar. 03, 2026. [Online]. Available: <https://nvidia-omniverse.github.io/PhysX/physx/5.1.1/docs/RigidBodyDynamics.html>
- [58] NVIDIA, “Articulations — PhysX 5.1.3 Documentation.” Accessed: Mar. 03, 2026. [Online]. Available: <https://nvidia-omniverse.github.io/PhysX/physx/5.1.3/docs/Articulations.html>
- [59] NVIDIA, “Tendon Visual Authoring — Omni Physics Documentation.” Accessed: Mar. 03, 2026. [Online]. Available: https://docs.omniverse.nvidia.com/kit/docs/omni_physics/latest/extensions/ux/source/omni.usdphysics.ui/docs/dev_guide/tendon_authoring.html
- [60] NVIDIA, “Rigid Body Collision — PhysX 5.1.0 Documentation.” Accessed: Mar. 03, 2026. [Online]. Available: <https://nvidia-omniverse.github.io/PhysX/physx/5.1.0/docs/RigidBodyCollision.html>
- [61] NVIDIA, “PxMaterial — PhysX 5.1.0 API Reference.” Accessed: Mar. 03, 2026. [Online]. Available: https://nvidia-omniverse.github.io/PhysX/physx/5.1.0/_build/physx/latest/class_px_material.html
- [62] M. Macklin *et al.*, “Small Steps in Physics Simulation.” Accessed: Mar. 03, 2026. [Online]. Available: <https://mmacklin.com/smallsteps.pdf>
- [63] NVIDIA, “Simulation — PhysX 5.4.1 Documentation.” Accessed: Mar. 03, 2026. [Online]. Available: <https://nvidia-omniverse.github.io/PhysX/physx/5.4.1/docs/Simulation.html>
- [64] NVIDIA, “Advanced Collision Detection — PhysX 5.1.2 Documentation.” Accessed: Mar. 03, 2026. [Online]. Available: <https://nvidia-omniverse.github.io/PhysX/physx/5.1.2/docs/AdvancedCollisionDetection.html>
- [65] NVIDIA, “Geometry — PhysX 5.4.1 Documentation.” Accessed: Mar. 03, 2026. [Online]. Available: <https://nvidia-omniverse.github.io/PhysX/physx/5.4.1/docs/Geometry.html>
- [66] V. Makoviyhuk *et al.*, “Isaac Gym: High Performance GPU Based Physics Simulation for Robot Learning,” in *NeurIPS 2021 Datasets and Benchmarks Track (Round 2)*, 2021. [Online]. Available: https://openreview.net/forum?id=fgFBtYgJQX_
- [67] NVIDIA, “What Is Isaac Sim?.” Accessed: Mar. 03, 2026. [Online]. Available: <https://docs.isaacsim.omniverse.nvidia.com/5.1.0/index.html>
- [68] NVIDIA *et al.*, “Isaac Lab: A GPU-Accelerated Simulation Framework for Multi-Modal Robot Learning.” [Online]. Available: <https://arxiv.org/abs/2511.04831>
- [69] NVIDIA, “GPU Rigid Bodies.” Accessed: May 02, 2026. [Online]. Available: <https://nvidia-omniverse.github.io/PhysX/physx/5.1.0/docs/GPURigidBodyBodies.html>
- [70] NVIDIA, “Isaac Sim: Extensions API.” Accessed: May 02, 2026. [Online]. Available: <https://docs.isaacsim.omniverse.nvidia.com/5.1.0/py/index.html>
- [71] M. Mittal *et al.*, “Orbit: A Unified Simulation Framework for Interactive Robot Learning Environments,” *IEEE Robotics and Automation Letters*, vol. 8, no. 6, pp. 3740–3747, 2023, doi: 10.1109/LRA.2023.3270034.
- [72] The Isaac Lab Project Developers, “Creating a Manager-Based RL Environment.” Accessed: Mar. 03, 2026. [Online]. Available: https://isaac-sim.github.io/IsaacLab/v1.0.0/source/tutorials/03_envs/create_manager_rl_env.html

- [73] The Isaac Lab Project Developers, “Multi-GPU and Multi-Node Training.” Accessed: Mar. 03, 2026. [Online]. Available: https://isaac-sim.github.io/IsaacLab/main/source/features/multi_gpu.html
- [74] E. Todorov, T. Erez, and Y. Tassa, “MuJoCo: A Physics Engine for Model-Based Control,” in *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2012, pp. 5026–5033. doi: [10.1109/IROS.2012.6386109](https://doi.org/10.1109/IROS.2012.6386109).
- [75] Bullet Physics Project, “Bullet Real-Time Physics Simulation: Home of Bullet and PyBullet.” Accessed: Apr. 22, 2026. [Online]. Available: <https://pybullet.org/wordpress/>
- [76] Pixar Animation Studios, “USD Frequently Asked Questions.” Accessed: Mar. 03, 2026. [Online]. Available: <https://openusd.org/release/usdfaq.html>
- [77] NVIDIA, “Physx Joint Schema — Omni Physics Documentation.” Accessed: Mar. 03, 2026. [Online]. Available: https://docs.omniverse.nvidia.com/kit/docs/omni_physics/latest/dev_guide/joints/physx_joint_schema.html
- [78] Open Robotics, “URDF (Unified Robot Description Format) — ROS 2 Documentation (Rolling).” Accessed: Mar. 03, 2026. [Online]. Available: <https://docs.ros.org/en/rolling/Tutorials/Intermediate/URDF/URDF-Main.html>
- [79] Open Robotics, “Adding Physical and Collision Properties to a URDF Model — ROS 2 Documentation (Foxy).” Accessed: Mar. 03, 2026. [Online]. Available: <https://docs.ros.org/en/foxy/Tutorials/Intermediate/URDF/Adding-Physical-and-Collision-Properties-to-a-URDF-Model.html>
- [80] The hardware_interface Contributors, “Joint Kinematics for ros2_control.” Accessed: Mar. 03, 2026. [Online]. Available: https://docs.ros.org/en/rolling/p/hardware_interface/doc/joints_userdoc.html
- [81] NVIDIA, “Tutorial: Import URDF — Isaac Sim Documentation (5.1.0).” Accessed: Mar. 03, 2026. [Online]. Available: https://docs.isaacsim.omniverse.nvidia.com/5.1.0/importer_exporter/import_urdf.html
- [82] The Isaac Lab Project Developers, “Importing a New Asset.” Accessed: Mar. 03, 2026. [Online]. Available: https://isaac-sim.github.io/IsaacLab/main/source/how-to/import_new_asset.html
- [83] NVIDIA, “Known Physics Limitations — Omni Physics Documentation.” Accessed: Mar. 03, 2026. [Online]. Available: https://docs.omniverse.nvidia.com/kit/docs/omni_physics/latest/dev_guide/guides/current_limitations.html
- [84] Y. Narang *et al.*, “Factory: Fast Contact for Robotic Assembly,” in *Robotics: Science and Systems (RSS)*, 2022. doi: [10.15607/RSS.2022.XVIII.065](https://doi.org/10.15607/RSS.2022.XVIII.065).
- [85] B. Tang *et al.*, “IndustReal: Transferring Contact-Rich Assembly Tasks from Simulation to Reality,” in *Robotics: Science and Systems (RSS)*, 2023. doi: [10.15607/RSS.2023.XIX.069](https://doi.org/10.15607/RSS.2023.XIX.069).
- [86] A. Allshire *et al.*, “Transferring Dexterous Manipulation from GPU Simulation to a Remote Real-World TriFinger,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2022, pp. 11802–11809. doi: [10.1109/IROS47612.2022.9981586](https://doi.org/10.1109/IROS47612.2022.9981586).

- [87] A. Handa *et al.*, “DeXtreme: Transfer of Agile In-hand Manipulation from Simulation to Reality,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2023, pp. 5977–5984. doi: [10.1109/ICRA48891.2023.10161417](https://doi.org/10.1109/ICRA48891.2023.10161417).
- [88] A. Serrano-Muñoz, D. Chrysostomou, S. Bøgh, and N. Arana-Arexolaleiba, “skrl: Modular and Configurable Library for Reinforcement Learning,” *Journal of Machine Learning Research*, vol. 24, no. 254, pp. 1–7, 2023, [Online]. Available: <https://jmlr.org/papers/v24/23-0112.html>
- [89] K. Or *et al.*, “Curriculum-reinforcement learning on simulation platform of tendon-driven high-degree of freedom underactuated manipulator,” *Frontiers in Robotics and AI*, vol. 10, p. 1066518, 2023, doi: [10.3389/frobt.2023.1066518](https://doi.org/10.3389/frobt.2023.1066518).
- [90] S. Guist *et al.*, “Safe & Accurate at Speed with Tendons: A Robot Arm for Exploring Dynamic Motion,” in *Robotics: Science and Systems (RSS) XX*, 2024.
- [91] R. Morimoto, M. Ikeda, R. Niiyama, and Y. Kuniyoshi, “Characterization of Continuum Robot Arms Under Reinforcement Learning and Derived Improvements,” *Frontiers in Robotics and AI*, vol. 9, p. 895388, 2022, doi: [10.3389/frobt.2022.895388](https://doi.org/10.3389/frobt.2022.895388).
- [92] R. Dhakate, T. Jantos, E. Allak, S. Weiss, and J. Steinbrener, “CaRoSaC: A Reinforcement Learning-Based Kinematic Control of Cable-Driven Parallel Robots by Addressing Cable Sag Through Simulation,” *IEEE Robotics and Automation Letters*, 2025, doi: [10.1109/LRA.2025.3564225](https://doi.org/10.1109/LRA.2025.3564225).
- [93] J. Garrido Campos, D. Silva-Muñiz, J. Rivera-Andrade, and E. Riveiro, “Integrated Simulation and Control Environment for the Development and Safe Start-up of Cable Driven Parallel Robots,” *International Journal of Computer Integrated Manufacturing*, vol. 38, no. 7, 2024, doi: [10.1080/0951192X.2024.2389640](https://doi.org/10.1080/0951192X.2024.2389640).
- [94] B. Fasquelle *et al.*, “Identification and Control of a 3-X Cable-Driven Manipulator Inspired From the Bird's Neck,” 2022, p. 11005. doi: [10.1115/1.4051521](https://doi.org/10.1115/1.4051521).
- [95] A. Hamon, Y. Aoustin, and S. Caro, “Cross Four-Bar Linkage for the Knees of a Planar Bipedal Robot,” in *2010 10th IEEE-RAS International Conference on Humanoid Robots (Humanoids)*, 2010. doi: [10.1109/ICHR.2010.5686851](https://doi.org/10.1109/ICHR.2010.5686851).
- [96] M. Zhang *et al.*, “Deep Reinforcement Learning for Tensegrity Robot Locomotion,” in *2017 IEEE International Conference on Robotics and Automation (ICRA)*, 2017, pp. 634–641.
- [97] N. Vahrenkamp and T. Asfour, “Representing the Robot's Workspace Through Constrained Manipulability Analysis,” *Autonomous Robots*, vol. 38, no. 1, pp. 17–30, 2015, doi: [10.1007/s10514-014-9388-z](https://doi.org/10.1007/s10514-014-9388-z).
- [98] A. Peidró, Ó. Reinoso, A. Gil, J. M. Marín, and L. Payá, “An Improved Monte Carlo Method Based on Gaussian Growth to Calculate the Workspace of Robots,” *Engineering Applications of Artificial Intelligence*, vol. 64, pp. 197–207, 2017, doi: [10.1016/j.engappai.2017.06.009](https://doi.org/10.1016/j.engappai.2017.06.009).